

Please use this PDF proof to check the layout of your article. If you would like any changes to be made to the layout, you can leave instructions in the online proofing interface. First, return to the online proofing interface by clicking "Edit" at the top page, then insert a Comment in the relevant location. Making your changes directly in the online proofing interface is the quickest, easiest way to correct and submit your proof.

Please note that changes made to the article in the online proofing interface will be added to the article before publication, but are not reflected in this PDF proof.

Review

Structural designs toward performance-balanced multifunctional batteries

Q1 Yang Mei,^{1,5} Xin Hu,^{1,5} Huirong Wang,^{1,5} Yongxin Huang,^{1,2,3,*} Li Li,^{1,2,3,4} Feng Wu,^{1,2,3,4} and Renjie Chen^{1,2,3,4,*}

¹Beijing Key Laboratory of Environmental Science and Engineering, School of Materials Science and Engineering, Beijing Institute of Technology, Beijing 100081, China

²Shandong Key Laboratory of Advanced Chemical Energy Storage and Intelligent Safety, Advanced Technology Research Institute, Beijing Institute of Technology, Jinan 250300, China

³Innovative Research Team in High-Safety Energy Storage System and Smart Microgrids of Guangdong Province, Beijing Institute of Technology (Zhuhai), Zhuhai 519088, China

⁴Collaborative Innovation Center of Electric Vehicles in Beijing, Beijing 100081, China

⁵These authors contributed equally

*Correspondence: huangyx@bit.edu.cn (Y.H.), chenrj@bit.edu.cn (R.C.)

<https://doi.org/10.1016/j.matt.2025.102469>

PROGRESS AND POTENTIAL Multifunctional batteries not only store energy but also bear loads, adapt to deformations, or fit into confined spaces, thereby enhancing their intelligence and applicability across diverse scenarios. Structural design has emerged as an effective strategy to improve battery functionality without altering the underlying chemical systems. This review analyzes structural design strategies such as component arrangement, interface continuity, topological optimization, and system integration used to develop representative micro-, deformable, integrated, and structural batteries. Key functional materials and advanced manufacturing techniques involved in these processes are also discussed. Furthermore, through scalable manufacturing, reliable integration, and robust packaging, batteries can be seamlessly incorporated as part of structural components, reducing system mass and volume and creating new opportunities in advanced architecture, transportation, and robotics.

The review also outlines design strategies for multifunctional battery modules and large-scale manufacturing as well as the application of machine learning to accelerate performance prediction and structural screening. However, challenges remain in areas such as electrochemical performance degradation, scalability in manufacturing, reparability, safety, and certification standards. A four-stage development road map illustrates the evolution of multifunctional batteries toward universality, multifunctionality, and intelligence, highlighting key materials, technologies, and performance metrics for each generation. Further advances in multifunctional batteries will accelerate the deployment of electric devices in various scenarios within future smart cities, contributing to more efficient and convenient intelligent living.

SUMMARY

Electric energy has propelled the development of clean transportation, intelligent buildings, and smart cities, while the diversification of application scenarios has put forward new requirements for its features. Therefore, the multifunctional batteries developed through structural design and integrated optimization have garnered significant interest due to their superior performance and good adaptability. Nevertheless, achieving a balance between battery performance and functionality continues to pose a challenge. This review examines structural design strategies for battery cells and systems to attain diverse functionality. The advanced materials and corresponding processing technologies employed in structural design have been summarized to provide feasible approaches. Subsequently, the representative architectures and corresponding characteristics of functional unit cells—involving microbatteries, deformable batteries, all-in-one batteries, and structural batteries—are thoroughly discussed based on classic cases. Attractively, a series of design principles have been proposed for the packaging and large-format integration of multifunctional batteries. Finally, the challenges and future prospects of multifunctional batteries, along with their design strategies, are discussed, culminating in a four-stage development road map.

Q4 Q3 Q2 INTRODUCTION

Secondary batteries represented by lithium-ion batteries (LIBs) have penetrated all aspects of society and have significantly changed people's lifestyles.^{1,2} The advancement of battery technology has been significant, particularly driven by the strategic inclusion of new energy vehicles and energy-storage industries across numerous nations.³ Nonetheless, as modernization and the smart society continue to evolve, intelligent equipment necessitates comprehensive demands on electrochemical energy-storage devices. These devices need to precisely adapt to a myriad of application scenarios (Figure 1A), including real-time human health monitoring, safe implantable medicine, underwater power supply with high mechanical load, and electric transportation with high space utilization, among others.^{4,5} Although the specific requirements for batteries vary across these scenarios and may involve complex factors such as biocompatibility or thermal stability under extreme conditions, which are beyond the scope of this review, they all require enhanced multifunctionality and/or tailored geometric configurations. Conventional batteries face significant limitations in these applications. Under dynamic deformation such as folding or stretching, battery components are prone to fracture and interface delamination, disrupting continuous electron and ion transport. During miniaturization, active material loading is reduced, while the risk of short circuits caused by fabrication errors increases, along with more severe localized heat accumulation. In addition, poor interfacial adhesion hinders the assembly of seamless all-in-one batteries suitable for flexible and miniaturized devices. Mechanical loading can also induce severe stress and thermal concentration, posing further challenges to battery safety (Figure 1B). These factors lead to suboptimal electrochemical and mechanical performance, including limited energy density, shortened cycle life, large bending radius, low ultimate strain, and reduced Young's modulus.

To address these limitations and enhance device-level functionality, various structural design strategies have been proposed and implemented. Common strategies in the structural design of multifunctional batteries include bionic architectures, compact configurations, topology optimization, and system-level integration, among others. Research on multifunctional batteries based on structural engineering can be traced back over two decades. In 2005, researchers employed carbon fiber (CF) as an auxiliary electrode material to develop batteries with compact shapes and potent environmental adaptation, marking an early recognition of the importance of structural design.⁶ To enhance the cruising range of hybrid vehicles, the storage team successfully integrated commercial LIBs into the automobile fairing, utilizing power supplied by integrated battery packs.⁷ Furthermore, Greenhalgh et al. proposed utilizing vehicle bodies themselves for energy storage, offering an alternative to conventional battery packs.^{7,8} These advances not only enhanced electrochemical performance but also drove innovation in battery architecture and integrated system architectures.

Batteries with specialized structures and functionalities often face intrinsic challenges in achieving a balance between competing design objectives.^{9,10} While structural innovations can enhance mechanical adaptability, integration, or precision,

they often come at the expense of electrochemical performance, manufacturability, or cost. These trade-offs are particularly pronounced in multifunctional batteries, where both energy storage and device-level mechanical or functional requirements must be met simultaneously. Four key dimensions have been identified where performance balancing is essential: functionality-electrochemical performance, structural complexity-processability, integration level-cell replaceability, and precision-cost. Improvements along one dimension often introduce challenges along the other. For example, improving mechanical flexibility may reduce energy density, increasing structural complexity may hinder scalable fabrication, and higher integration levels may reduce maintenance accessibility. Additionally, in practical battery development, the balance between functionality and electrochemical performance, as well as structural complexity and processability, is currently of highest concern, as they directly affect the feasibility and efficiency of multifunctional battery prototypes. In contrast, integration-replaceability and precision-cost trade-offs tend to emerge at later stages of engineering or system-level deployment. Advances in material science, structural design, and manufacturing techniques are expected to mitigate these trade-offs and enable high-performance multifunctional batteries.^{11,12}

This review focuses on multidimensional structural design strategies for performance-balanced multifunctional batteries. The structural designs of individual cells enable their suitability of power supply for special thermodynamic and kinetic properties in various scenarios while addressing issues related to aesthetic compatibility and device integration. The structural design of battery modules can enhance the correlation among components and address issues in system performance optimization and multifunction integration. Ultimately, upgrading or reconstructing structural batteries using existing batteries is feasible by customizing materials and structures. These innovative structural designs have considerably enhanced the performance and functionality of energy devices. More importantly, such customizable structure greatly improves the compatibility between batteries and devices, showcasing enhanced adaptability in constrained spaces and extreme environments. Based on the above understanding, we summarize the development prospects and challenges of structural designs for functional batteries and discuss their future development directions. This review aims to provide structural design strategies and related supporting technologies, serving as a design guideline for the development of advanced multifunctional power systems across various battery chemistries.

MECHANICAL-ELECTROCHEMICAL ANALYSIS OF FAILURE MODES

Battery components will undergo plastic deformation after being subjected to external forces or prolonged high-load operation, leading to device failure. To better analyze the failure mechanism during battery deformation and propose improvement strategies from the perspective of structural design, the failure modes of functional batteries are specifically analyzed here in conjunction with the mechanical-electrochemical coupling formula.

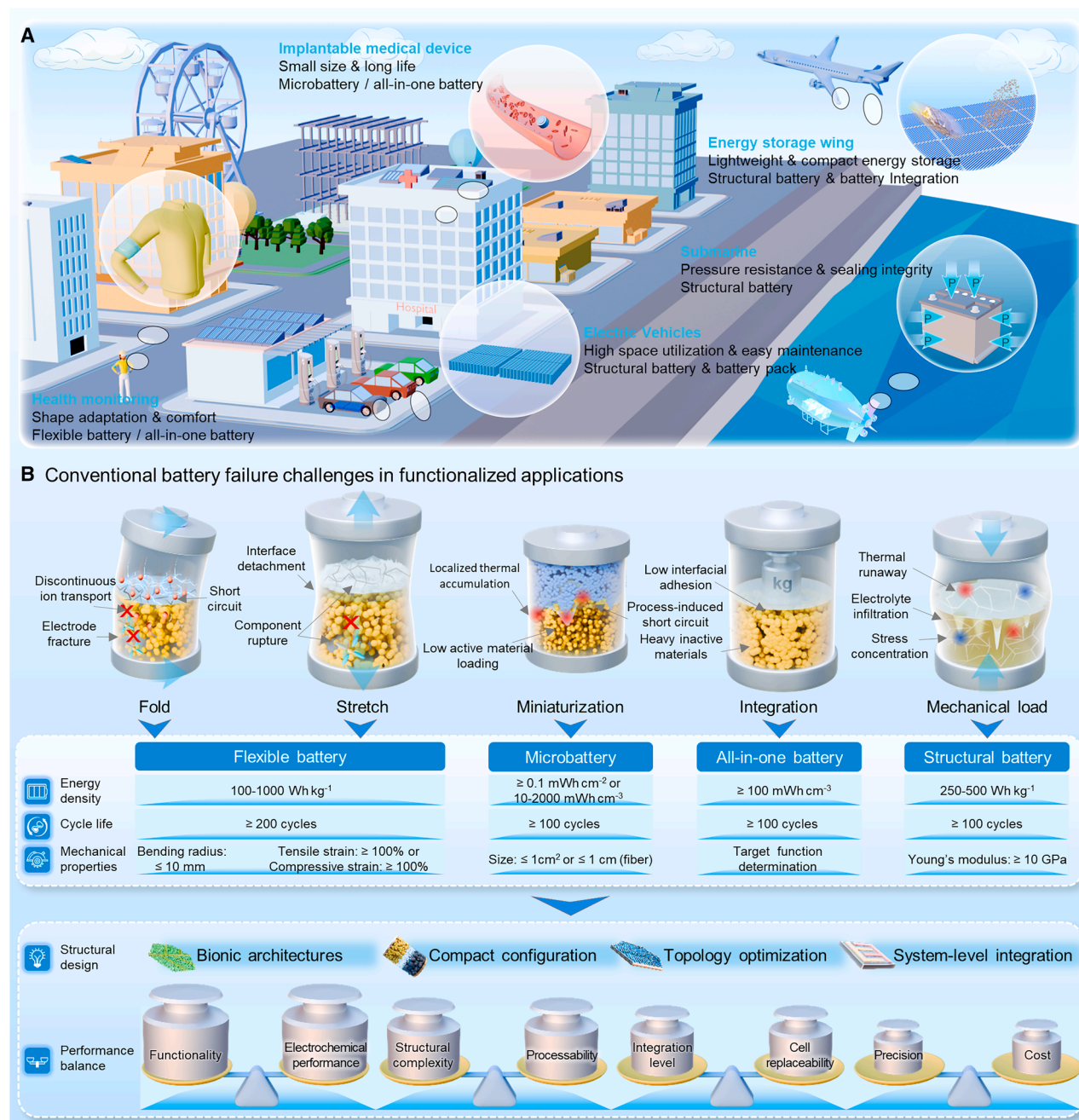


Figure 1. Multifunctional battery application demands and challenges

(A) Representative multifunctional battery applications and associated performance requirements.

(B) Degradation challenges of conventional batteries in functionalized applications and consequent multifunctional battery demands, structural design strategies, and performance balance challenges.

As shown in the mechanism diagram of the second row in Figure 2, when the battery is subjected to an external force impact, its internal structure will undergo inertial displacement and deformation. The relative displacement between different components forms a shear effect, generating internal shear stress. The failure mode due to shear stress-electrochemical coupling can be analyzed with reference to Equation 1¹³:

$$\partial c / \partial t = \nabla \cdot (D \nabla c - D c \Omega / RT \cdot \nabla \sigma_{\text{shear}}), \quad (\text{Equation 1})$$

where c is the Li^+ concentration, D is the Li^+ diffusion coefficient, Ω is the partial molar volume of Li^+ , σ_{shear} is the shear stress component, R is the gas constant, and T is the absolute temperature. Shear stress can damage the electrode or separator

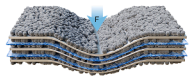
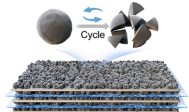
Failure mode	Mechanical-Electrochemical coupling formula	Decisive parameter	Coding strategy	Failure schematic diagram
Shear	$\partial c / \partial t = \nabla \cdot (D \nabla c - D c \Omega / RT \cdot \nabla \sigma_{\text{shear}})$	D	3D mesh binder; Multistage modified separator	
Tension and Compression	$\sigma = C / (\varepsilon - (c - c_0)) I \cdot \Omega / 3)$	σ	Structural reinforcing materials; Gradient pore electrode	
Load-bearing	$R_{\text{contact}} = 1 / (\sigma_c \cdot A(\sigma_{\text{comp}}))$	$A(\sigma_{\text{comp}})$	Micro-arch support structure; Load-bearing cell design	
Fatigue	$dD/dN = C(\Delta \varepsilon)^m$	$\Delta \varepsilon$	Self-healing binder; Shape memory alloy	

Figure 2. Schematic diagrams of the battery under various failure modes and their corresponding mechanical-electrochemical coupling formulas, decisive parameters, and coding strategies

structure (e.g., pore closure and separator bending), reduce D, and exacerbate the concentration polarization, which in turn triggers cell failure due to dendrite/lithium precipitation. For the failure behavior caused by shear stress, damage resistance protection can be achieved by coding strategies of three-dimensional (3D) mesh binder and multistage pore diaphragm. The structural design of 3D mesh binder can dissipate energy through network reconstruction during shear deformation and inhibit the active material from falling off,¹⁴ while the multistage porous diaphragm can dissipate shear stresses by absorbing the impact energy through the ceramic coating on the surface and the polymer matrix on the bottom layer to maintain the integrity of the battery components.¹⁵

When the cell is subjected to non-uniform shocks or extreme temperature changes, the internal components will undergo stretching behavior, as shown in the mechanism diagram of the second row in Figure 2. The failure mode due to stretch-electrochemical coupling can be analyzed by referring to Equation 2. This equation is also applicable to the compression-electrochemical coupling failure mode when the device is subjected to compression or impact.^{13,16}

$$\sigma = C / (\varepsilon - (c - c_0)) I \cdot \Omega / 3), \quad (\text{Equation 2})$$

where σ is the stress tensor (tensile/compressive stress), C is the elastic stiffness matrix (material stiffness properties), ε is the strain tensor (caused by the change in lithium concentration), c_0 is the initial Li^+ concentration, and I is the unit tensor. External tensile or compressive forces are directly superimposed on the diffusion-induced stresses, exacerbating the local stress concentration. When the total stress σ exceeds the material strength, the component will sprout cracks and expand (e.g.,

destruction of electrode and solid electrolyte interphase), which in turn leads to the failure of the cell. For the failure behavior due to stretching/compression, damage resistance protection can be achieved through the encoding strategy of structural reinforcement material composite and gradient pore electrode. The composite of structural reinforcement materials (e.g., reinforced CFs) can improve the toughness of the component and enhance the tensile resistance.¹⁷ while the buffer layer of gradient pore electrodes can absorb the pressure and improve the compression resistance to maintain the normal operation of the device under compression.¹⁸

When the battery is subjected to static long-term load bearing or dynamic impact load bearing, its internal structure will experience structural creep or transient compression collapse, leading to failure. The failure mode due to load bearing-electrochemical coupling can be analyzed by referring to Equation 3¹⁹:

$$R_{\text{contact}} = 1 / (\sigma_c \cdot A(\sigma_{\text{comp}})), \quad (\text{Equation 3})$$

where R_{contact} is the contact resistance, σ_c is the material conductivity, σ_{comp} is the compressive stress, and $A(\sigma_{\text{comp}})$ is the effective contact area. Appropriate load bearing will increase the contact area A and reduce R_{contact} . However, long-term static load bearing or instantaneous impact load bearing will crush the electrode, leading to battery failure, as shown in the mechanism diagram of the fourth row in Figure 2. To address the failure behavior caused by load bearing, damage protection can be achieved by coding strategies for the design of micro-arch support structures and load-bearing batteries (body-battery integrated design). The micro-arch support structure disperses the pressure to the support points to avoid localized collapse,²⁰ while the load-bearing battery design disperses the

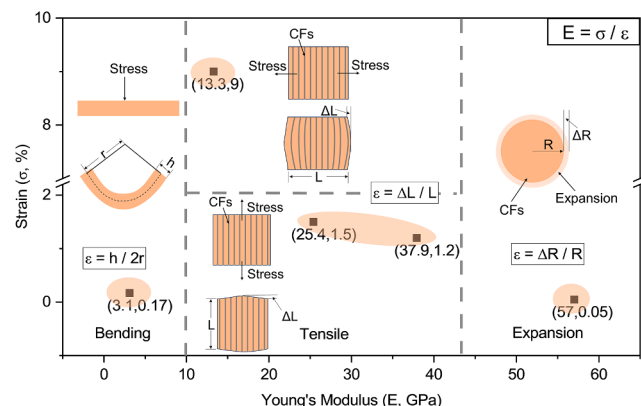


Figure 3. CF materials with different moduli employed in different mechanical situations

From Asp et al.,³⁰ Bouton et al.,³¹ Carlstedt et al.,³² and Jin et al.¹⁷¹

local load-bearing pressure to the whole, improving the battery's ability to resist load-bearing failure.²¹

When the battery is used under long-term conditions such as high-rate charging and discharging, low-temperature charging, and large temperature differences between day and night, failure behaviors such as electrode material detachment, active particle pulverization, and current collector fatigue fracture will occur inside it, as shown in the schematic diagram of the fifth mechanism in Figure 2. The failure mode due to fatigue-electrochemical coupling can be analyzed with reference to Equation 4²²:

$$dD/dN = C(\Delta\epsilon)^m, \quad (\text{Equation 4})$$

where D is the damage variable, N is the number of cycles, $\Delta\epsilon$ is the cyclic strain amplitude (caused by volume change during charging and discharging), and C and m are the fatigue parameters of the material. The larger the cyclic strain amplitude $\Delta\epsilon$, the faster the battery damage accumulates and aggravates the battery failure. To address the failure behavior due to fatigue, damage protection can be achieved by encoding strategies of self-healing binders and shape-memory alloy coatings. The self-healing binder can enhance the adhesion between the electrode and the collector based on the dynamic bond reorganization function to maintain good electronic conduction,²³ while the shape-memory alloys can adaptively repair damage to enhance the fatigue life.²⁴

Based on the mechanism analysis of the above failure modes, the mechanical-electrochemical coupling formulas, decisive parameters, and corresponding coding strategies of the battery under each failure mode are shown in Figure 2.

KEY CF MATERIAL AND PROCESSING TECHNOLOGIES FOR MULTIFUNCTIONAL BATTERIES

The electrochemical energy-storage systems generally have a well-defined composition and structure, including active components (electrodes) and inactive components (electrolytes, separators, current collectors, and encapsulants). Thus, effectively

integrating high energy density with multifunctional design within the limited volume of conventional batteries remains challenging.^{25,26} To address this, the support of multifunctional materials, such as CF, is necessary for the versatility of structurally designed functional batteries. This section will therefore introduce CF as a representative functional material and outline multiscale processing technologies.

Key CF materials for structural designs

Carbon materials are widely employed in structurally designed batteries due to their outstanding electrical and mechanical properties as well as their highly tunable architectures.^{27,28} CFs play a pivotal role in structural batteries, enabling them to withstand various mechanical stresses. They are widely integrated into composite anodes and cathodes to enhance the mechanical robustness of the battery under tensile, bending, and impact forces. In addition to external mechanical stimuli, internal expansion of the electrodes during electrochemical cycling can destabilize the electrode-electrolyte interphase (EEI), which critically affects the long-term electrochemical stability of structural batteries. Therefore, CFs are required not only to accommodate external mechanical loads but also to adapt to expansion-induced strain to ensure stable cycling behavior.

As summarized in Figure 3, CFs are designed to address mechanical challenges under various application scenarios. In flexible batteries, for example, electrodes must accommodate bending deformation to adapt to the articulation of soft robotic joints. The corresponding strain under structural bending can be expressed as

$$\epsilon = h/2r, \quad (\text{Equation 5})$$

where h is the thickness of the structural material and r is the bending radius.²⁹ A smaller r , as required for high curvature in flexible applications, results in larger strain and, consequently, greater mechanical stress. According to the definition of Young's modulus,

$$E = \sigma/\epsilon, \quad (\text{Equation 6})$$

where E is Young's modulus, σ is the stress, and ϵ is the strain. CFs with lower moduli are preferred for high-bending scenarios to prevent structural damage.

In contrast, when CFs are used as structural skins in rigid electric devices such as the wings of electric aircraft or the chassis of electric vehicles, they must endure substantial transverse or longitudinal tensile loads. The tensile strain in such cases can be expressed as

$$\epsilon = \Delta L/L, \quad (\text{Equation 7})$$

where ΔL is the elongation under tensile force, and L is the original length of the material. To ensure minimal deformation under stress, composite structural electrodes must exhibit small ΔL , implying a requirement for high-modulus CFs. For instance,

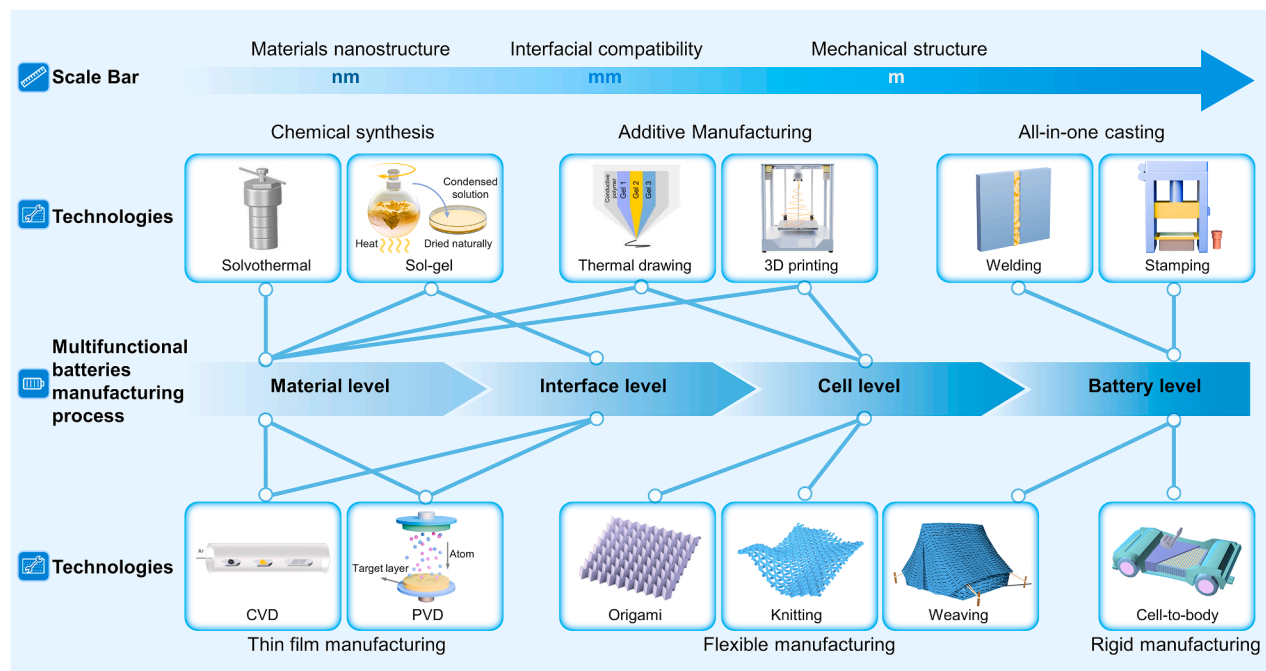


Figure 4. Summary of typical methods at different scales used in processing functional batteries

reinforced CF-based electrodes provide 13.3 GPa in transverse direction with strain of 9% and 25.4 GPa in longitudinal direction with strain of 1.5%.^{30,31} As illustrated in Figure 3, mechanical deformation differs significantly along various orientations of CFs, indicating strong anisotropy in their mechanical response. Future design and fabrication of structural batteries should leverage the superior longitudinal mechanical strength of CFs to meet structural integrity requirements.

Beyond mechanical support, the preservation of electrochemical performance during cycling is critically dependent on the stability of the EEI. It is well established that the EEI formed during the initial cycles of LIBs passivates the electrode surface, preventing continuous consumption of active species. However, brittle EEI layers are prone to rupture and repeated reconstruction due to volumetric expansion, leading to the fading of accelerated capacity. Maintaining a stable microstructure of the electrode is thus essential for long-term performance. The radial strain induced by lithium intercalation in cylindrical CFs is described by

$$\varepsilon = \Delta R/R, \quad (\text{Equation 8})$$

where ΔR is the increase in fiber radius due to lithium insertion, and R is the original fiber radius. To suppress such expansion, CFs embedded in structural battery electrolyte with high elastic moduli (such as those with moduli up to 57 GPa) are essential.³²

In summary, CFs serve as versatile structural materials capable of addressing a broad range of mechanical challenges in functional batteries. Their tunable mechanical properties offer a rational basis for material selection in composite electrode

design across diverse operational environments, supporting both structural resilience and electrochemical stability.

Processing technologies for structural designs

With the increasing demand for high-performance and multifunctional energy-storage devices, structural design across different dimensional scales has emerged as a critical approach to tailor the physical, mechanical, and electrochemical properties of batteries. Structures ranging from zero-dimensional (0D) to four-dimensional (4D) play unique roles in enhancing energy-storage performance, enabling applications such as compact integration, flexible/wearable electronics, and responsive smart systems. For instance, 0D and one-dimensional (1D) structures can offer high surface areas and fast ion/electron transport paths, two-dimensional (2D) films provide conformal interfaces for coating electrodes, 3D architectures enable high-loading active materials, and 4D structures (those capable of dynamic shape transformation) respond to environmental stimuli, showing great promise for adaptive systems.

Correspondingly, various processing technologies have been developed to fabricate components and architectures across multiple levels: 0D quantum dots and 1D nanowires at the material level, 2D thin films/coatings at the interface level, 3D spatial frameworks at the cell level, and 4D smart architectures at the battery level. In Figure 4, the conventional technical methods for structural designs are systematically summarized according to different scales and the corresponding technical characteristics analyzed. Among them, chemical synthesis (including solid-phase reaction, sol-gel method, solvothermal method, and chemical vapor deposition [CVD]) is the most basic means of material preparation, which can achieve fine structural

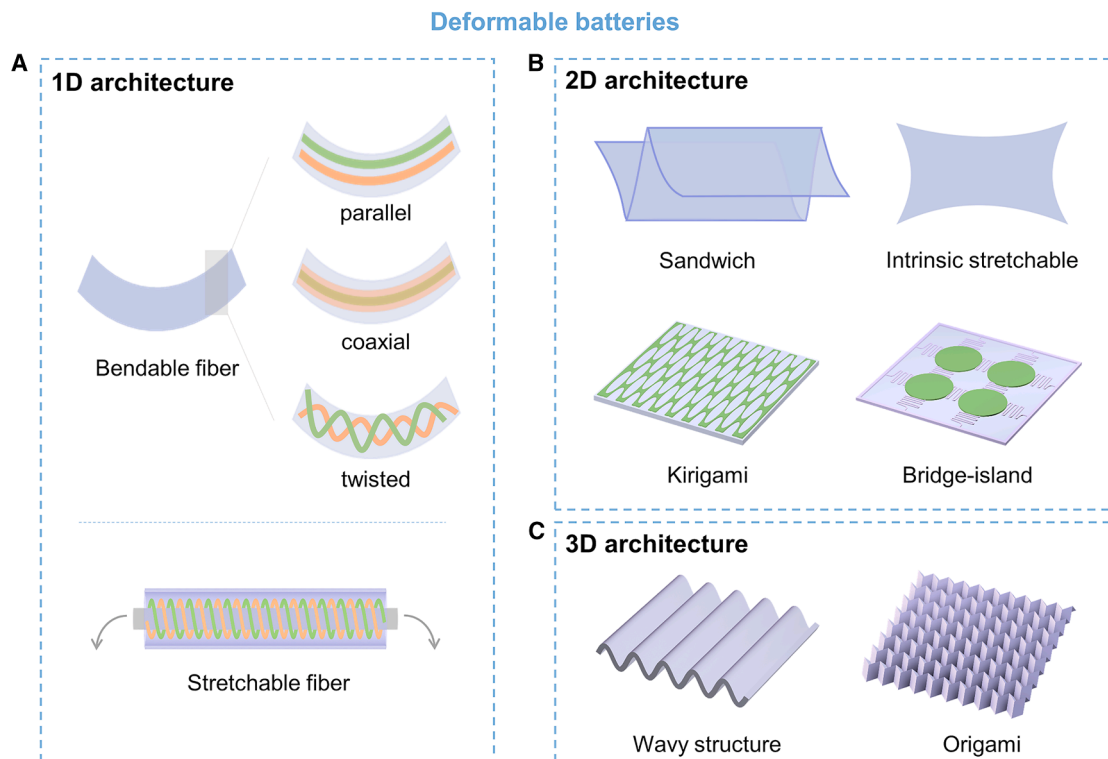


Figure 5. Structural engineering of deformable batteries across different dimensions

(A) 1D architecture: bendable (parallel,^{43–45} coaxial,^{46–48} twisted^{49,68}) and stretchable.^{74,75}
 (B) 2D architecture: sandwich,^{52,53} intrinsic stretchable,^{54–57} kirigami,^{58–60} and bridge-island.^{61,62}
 (C) 3D architecture: wavy³³ and origami.⁶⁷

control as a preparation method at present. Physical vapor deposition (including pulsed laser deposition, magnetron sputtering, and vacuum coating) has the advantages of uniform film formation and high precision, but the expensive equipment limits large-scale production and large-scale application.³³ Additive manufacturing technology (including thermal drawing, 3D/4D printing, and inkjet printing), as an emerging 2D/3D/4D manufacturing process, can usually meet the flexibility requirements for the preparation of irregular devices, but the configuration of precursor solutions is complex.³⁴ This technology offers unique benefits such as efficient use of space within battery systems and the ability to respond to material deformation. In the preparation of flexible/deformable structural design batteries, processes such as origami, knitting, and weaving enable effective integration of devices, but the balance of mechanical and electrochemical properties remains a challenge.^{35,36} Among the various fabrication methods, 4D printing technology stands out due to its ability to efficiently utilize the internal space of the batteries and to respond to material deformation in a timely manner.³⁷ For large-scale battery pack production, smarter integrated casting and higher precision laser welding are booming, which can improve the production quality and efficiency to enable customized manufacturing of special structures.³⁸ Furthermore, the compatibility of materials with devices and circuits should be considered comprehensively, in addition to the improvement of certain parameters. To overcome the challenges

in design, materials, and preparation of functional batteries, the combined efforts of materials science, information science, and mechanics should be achieved in the future.

STRUCTURAL DESIGNS FOR FUNCTIONAL UNIT CELLS

In the structural design of unit cells, functional physical characteristics are incorporated to accommodate various application scenarios and continuously evolving electronic devices. The physical properties of batteries, such as deformability, size, geometry, and strength, usually determine their potential implementation scenarios before the required electrochemical performance. In this section, structural design strategies and component integration methods will be introduced for representative unit functional cells, including deformable batteries, microbatteries, all-in-one batteries, and structural batteries.

Deformable batteries

Deformable batteries are widely used in wearable electronics, implantable medical devices, and smart home systems due to their advantages in adapting to motion and conforming to irregular device surfaces.^{39–42} Flexible fiber batteries, plane batteries, and 3D batteries with various integrated structures are the basic types of device-level deformable batteries; the corresponding schematic diagrams are shown in Figure 5. The descriptions of deformable batteries with various structures, including battery

components, electrochemical properties, deformability, and application demonstrations, are summarized in Table 1.^{43–67}

Fibrous batteries are usually bendable and weavable due to their linear structures and smaller diameters, making them highly applicable to wearable electronics. The cathodes and anodes in the fibrous batteries can be integrated in parallel,^{43–45} coaxial,^{46–48} and twisted configurations.^{49,68} In addition, stretchable fiber batteries can also be obtained through structural design with helix structure or pre-stretching methods to accommodate tensile strain besides using the intrinsic stretchable fiber materials (Figure 5A).^{50,51} Different structural designs have significant impact on the performance of the batteries. Overall, four architectures of fiber batteries exhibit different benefits and drawbacks in terms of energy density, durability, processability, applicability, stretchability, and kinetics. (1) Due to the long ion-migration path, fiber batteries usually suffer from high internal resistance and poor kinetic properties. In particular, the parallel fiber batteries with smaller electrode contact areas and larger diameters generally exhibit inferior kinetic performance. (2) The manufacture of fiber batteries faces many difficulties compared to conventional batteries, including the high cost of gel electrolyte, uneven distribution of active material, low interfacial adhesion, and the complexity of the preparation process. (3) Since the positive and negative electrodes can be processed independently, the parallel and twisted fiber batteries are relatively easy to process. Besides, the twisted fiber batteries are similar in structure to fabric, thus being suitable for wearable electronic devices to provide superior comfort and process compatibility. (4) Coaxial fiber batteries have tight interfacial contact and small free space, exhibiting the advantages of high loading and superior volumetric energy density. (5) Stretchable fiber batteries with more dimensional deformability provide the advantage of high applicability. However, the introduction of elastic inactive material and low electrode space occupancy in the stretched state reduces its weight and volume energy density.

Various processing methods and electrode material strategies have been explored to enhance the electrochemical performance, mechanical flexibility, and scalability of deformable fiber batteries.⁶⁹ These strategies are often closely related to the underlying structural design of the batteries.

Thermal drawing has emerged as a scalable method for producing arbitrarily long, fully functional fiber batteries with parallel-aligned components (Figure 6A).^{70–72} With this method, polyvinylidene fluoride (PVDF)-based gel electrodes and electrolytes with tunable rheological properties are heated and then drawn into fibers under a nitrogen atmosphere. The viscoelastic fluid-fluid consolidation process ensures parallel alignment of components, enabling efficient ion transport. By adjusting the capstan speed, fiber dimensions can be precisely tailored without altering the pre-form. A 140-m-long fiber battery fabricated via this approach demonstrates a discharge energy of 15.5 mWh cm^{−1}, suitable for applications in submarine drones, LiFi fabrics, and aerial communications.

To address processing complexity and cost, coaxial zinc ion battery (ZIB) fibers utilize a composite hollow fiber soaked in KOH electrolyte (Figure 6B).⁴⁴ The spandex-polyester hybrid substrate exhibits high water absorption, low internal resistance,

and flexibility, meeting the demands of wearable energy storage. However, challenges remain in optimizing encapsulation methods and long-term electrolyte retention.

3D printing enables the fabrication of twisted LIB fibers with enhanced design freedom and scalability (Figure 6C).^{46,68,73} The cathode and anode inks are extruded into fibers, followed by twisting with a PVDF-co-hexafluoropropylene (HFP)-based gel electrolyte. The resulting battery achieves a specific capacity of 110 mAh g^{−1} at 50 mA g^{−1} and maintains stable operation under bending, as evidenced by uninterrupted light-emitting diode (LED) illumination during mechanical deformation. This highly integrated twisted pair configuration ensures good flexibility and stability.

For wearable devices that require high elasticity (e.g., electronic skin and artificial muscles), stretchable fiber batteries have been developed using intrinsically elastic components and a helical structural design.^{74,75} A Ga-In liquid metal anode paired with a polyacrylic acid (PAA)-based gel electrolyte and CF@Pt cathode achieves highly elastic and stretchability (Figure 6D).⁵¹ In addition, pre-stretched elastic rods wrapped with electrode fibers and coated with gel electrolyte self-assemble into helical batteries upon relaxation, maintaining stable structural and electrochemical performance under repeated tensile strains.^{49,50}

The reduction in fiber diameter can significantly improve the comfort of the woven products and is necessary for wearable electronic devices; however, the ensuing dramatic increased processing difficulties and weakened electrochemical properties pose challenges for superior fiber batteries.^{76–78} The practical viability of some manufactured yarns is contested, since the diameter of a deformable fiber battery is not explicitly limited by current testing standards. Furthermore, most of the reported deformable fiber batteries are finite in length, and only a few efforts are currently available to achieve meter-scale fabrication, so it is meaningful for optimizing industrial processing methods to obtain scalable fabrication.^{79,80}

For planar 2D deformable batteries, four architectures are generally implemented, namely the sandwich, intrinsic flexible, kirigami, and bridge-island (Figure 5B). Each structure features distinct design principles and trade-offs in terms of processability, mechanical durability, electrochemical performance, and applicability.

The sandwich configuration closely resembles traditional battery architectures, where electrodes, electrolytes, separators, current collectors, and packaging are stacked in layers between flexible substrates.^{52,53} This structural similarity to coin cells ensures compatibility with mature fabrication techniques, offering excellent processability and scalability. Moreover, compared to fiber batteries, sandwich-type designs possess shorter ion and electron transport distances, resulting in superior kinetic performance. To achieve mechanical flexibility, carbon-based materials such as carbon nanotube (CNT) films, graphene aerogels, and MXene films are often used as flexible electrodes due to their excellent conductivity and mechanical compliance.^{81,82}

Different paper electrodes or current collectors have been prepared by CVD, electrospinning, suction filtration, self-assembly, printing, and other methods to support active materials. Additionally, thin metal foils may serve as flexible current collectors.

Table 1. Summary of recent deformable batteries reported in literature

Configuration	Cathode	Anode	Electrolyte@ separator	Energy density/ capacity/ power density	Deformability evaluation	Deformation	Application	Reference
1D@parallel fiber	LCO@Ni-carbon textile	graphite@ Cu-carbon textile	LiPF ₆ /EC/DMC@ Celgard	329 Wh L ⁻¹	~90% CR over 150 charge-discharge cycles with 2,000 bending cycles at 4.0 mm	flexible	LED screen	Luo et al. ⁴³
	LFP	LTO	LiTFSI/PC/EC-PVDF gel	123 mAh	after 1,000 bending cycles, 96% CR over 30 charge-discharge cycles	flexible	LiFi fabric sample	Zhang et al. ⁴⁴
	CNT/H _{0.82} MoO _{3.26} fiber	Zn wire	Zn(CF ₃ SO ₃) ₂ @GF	18.9 mWh cm ⁻³	91% CR after bending 3,500 times	flexible	sensor/smart watch/timer	Li et al. ⁴⁵
1D@coaxial fiber	commercial air electrode	Zn wire	KOH@hollow composite fiber	754 mAh g ⁻¹ Zn at 1 mA cm ⁻¹	stable operation voltage after folding, bending, twisting, and recycling	flexible	LED/electronic watch/smart phone	Wang et al. ⁴⁶
	V-Co ₃ O ₄ @ carbon cloth	Zn spring	KOH-PVA gel	455 W h L ⁻¹ at 7 mA cm ⁻³	stable voltage when it is stretched, bent, or twisted 100 times	flexible/ stretchable	LED screen	Rao et al. ⁴⁷
	LCO@CMF	LTO@CMF	LiPF ₆ /DEC/EC@ Celgard	215 mWh cm ⁻³ at 1 C	similar performance before and after 100 twisting/releasing cycles	flexible	LEDs	Wu et al. ⁴⁸
1D@twisted fiber	LFP	LTO	LiPF ₆ /DEC/EC-PVDF-co-HFP gel	110 mAh g ⁻¹ at 50 mA g ⁻¹	stable operation after bending	flexible	LED/fabric integration	Wang et al. ⁴⁹
1D@stretchable fiber@helix	graphene-polyaniline @SF	Zn@SF	ZnSO ₄ -PVA gel	36.04 mWh cm ⁻³ at 0.003 W cm ⁻³	71% CR after being stretched 900%	stretchable	LED screen	Zhao et al. ⁵⁰
1D@stretchable fiber@enwinded	MoS ₂ @CNT fiber	Li-ZnO@ CNT fiber	LiTFSI-PEO gel	1132 mAh g ⁻¹ at 50 mA g ⁻¹	90.3% CR after being stretched at 100% strain for 100 times	stretchable	NR	Liu et al. ⁵¹
2D@sandwich	S/TG	Li foil	DOL/DME/LiTFSI@ Celgard	243 Wh kg ⁻¹ at 0.2 C	LEDs at 0° and 180° positions	flexible	LED screen	Senthil et al. ⁵²
	Co ₃ O ₄ @carbon cloth	Zn@carbon cloth	CCNF-PDIL SSEs	775 mAh g ⁻¹ at 2 mA cm ⁻¹	stable voltage plateaus under different bending states of 0°, 45°, 90°, and 180°	flexible	thermohygrometer	Xu et al. ⁵³

(Continued on next page)

Table 1. Continued

Configuration	Cathode	Anode	Electrolyte@ separator	Energy density/ capacity/ power density	Deformability evaluation	Deformation	Application	Reference
2D@intrinsic stretchable	Ag ₂ O	Zn	KOH-PVA gel	NR	1,500 cycles of 20% uniaxial stretching	stretchable	epidermal sweat-sensing platform	Yin et al. ⁵⁴
	LiMn ₂ O ₄ @ Ag composite membrane	V ₂ O ₅ @Ag composite membrane	LITFSI-PAM gel	60 mAh g ⁻¹ at 1 C	65% CR at 50% strain	stretchable	NR	Chen et al. ⁵⁵
	MnO ₂ @ PUS/AgNW	Zn@PUS/A gNW	KOH/ZnO-PAA gel	3.6 mAh·cm ⁻²	89% CR at 100% strain	stretchable	LED	Zhu et al. ⁵⁶
	VOPO ₄ @ PDMS/rGO sponge	hard carbon@ PDMS/rGO sponge	NaClO ₄ /PC-P(VDF-HFP) gel	103 mAh g ⁻¹ at 0.1 C	89% CR after 100 cycles of stretching to 50% strain	stretchable	LED	Li et al. ⁵⁷
2D@kirigami	MoO ₃ @Mo foil	Mg foil	calcium alginate gel	196 μW cm ⁻² at 200 μA cm ⁻²	35% reversible elasticity for uniaxial extension and 20% for biaxial extension	stretchable	LED/on-skin biomedical sensor patch	Karami-Mosammam et al. ⁵⁸
	Ni@PUS	Zn@PUS	KOH-PAA gel	19.7 mW cm ⁻² at 29 mA cm ⁻²	stable cycling profiles under 0%, 50%, and 100% strain	stretchable	small fan	Qu et al. ⁵⁹
	carbon black	Al foil	NaCl@wiper tissue	NR	stable operation in an expanded state (a strain of 72%)	stretchable	LED	Choi et al. ⁶⁰
2D@bridge-island	carbon cloth/ NdDCF-OIM/ Co-800	Zn foil	KOH/Zn (CH ₂ COO) ₂ -PAM-co-PAA gel	NR	75% peak power retention after 600 tensile cycles	stretchable	electronic watch	Zhao et al. ⁶¹
	LGO	LTO	LiClO ₄ /EC/DMC-PEO gel	1.1 mAh cm ⁻² at 0.5 C	similar capacity densities under 300% strain	stretchable	LED	Xu et al. ⁶²
3D@wavy	Ni@Cu-spandex textiles	Zn@Cu-spandex textiles	KOH-CMC-PAA gel	31.2 mW cm ⁻²	maximum stretchability of 100% with good conductivity	stretchable	attitude sensor	Qu et al. ⁶³
	LFP@C nanofibers	SnO ₂ @C nanofibers	LiPF ₆ /EC/DEC-PVDF-HFP gel	138.6 mAh g ⁻¹ at 0.5 C	98.9% CR in stretched states after 5 cycles of 30% strain and release	stretchable	LED	Kwon et al. ⁶⁴

(Continued on next page)

Table 1. Continued

Configuration	Cathode	Anode	Electrolyte@ separator	Energy density/ capacity/ power density	Deformability evaluation	Deformation	Application	Reference
3D@origami	NiCo@Au/CNT paper	Zn@Au/CNT paper	KOH/Zn (CH ₃ COO) ₂ - PANa gel	53.7 mAh g ⁻¹ at 23 A g ⁻¹	87% and 97% CR, respectively, after 500 stretching cycles and 1,500 compression cycles	stretchable	LED	Liu et al. ⁶⁵
	Fe ₃ O ₄	Zn	KOH@cellulose chromatography paper	NR	folding process demonstration	flexible	NR	Gonzalez-Guerrero and Gomez ⁶⁶
	LCO@CNT-coated paper	LTO	LiPF ₆ /EC/DMC/DEC@Celgard	0.2 mAh cm ⁻² (unfolded), 1.0 mAh cm ⁻² (compressed and cycled) at 20 mA g ⁻¹	1,340% linear deformability	flexible	LED	Song et al. ⁶⁷

TG, tragacanth gum; NR, not reported; CR, capacity retention; CMF, carbon nanotube woven microfilm; SF, silver fiber; PUS, polyurethane sponge; PAA, polyacrylic acid; PANa, sodium polyacrylate; AgNW, Ag nanowire; PDMS, polydimethylsiloxane; NDDCF-OIM/Co, nitrogen-doped dodecahedral carbon framework of ordered interconnected macropores loaded with monodisperse cobalt nanoparticles; CMF, carbon nanotube woven macrofilm.

Q8

Q9

For example, a flexible lithium metal battery with a sandwich architecture was reported using an ultrathin MXene/CNT composite anode and a LiFePO₄-based cathode via spin coating (Figure 7A),⁸³ exhibiting improved electrochemical stability and reversible capacity. Nevertheless, its performance under repeated mechanical deformation was not assessed. Despite their structural advantages, most sandwich batteries are non-stretchable, as many constituent materials lack sufficient tensile strength or elasticity to withstand large deformations.

Intrinsically flexible planar batteries are designed by integrating stretchable components to achieve comprehensive mechanical flexibility.^{54–57} These components are typically arranged in sandwich or interdigital configurations, enabling the battery to undergo significant elongation and thus making it well suited for applications such as soft electronics and wearable devices. To construct such systems, stretchable electrodes can be prepared by applying ultra-thin active material films over the polymer-based collectors.⁵⁵ Based on this strategy, a film-type stretchable rechargeable battery was developed using a composite current collector composed of Au nanosheets, Ag flakes, and copolymer, with spray-coated active materials and separators (Figure 7B).⁸⁴ The three-layer structure was bonded into a unified stretchable device, which served as a substrate for directly printing bilayer circuit lines. The integrated system demonstrated stable operation of LEDs under 125% biaxial strain under water, highlighting its mechanical and electrical reliability. Alternatively, intrinsically stretchable electrodes can be fabricated by blending active materials with polymers and regulating the crosslinking process. For example, PVDF-based organogels with controlled crystal growth have been used to generate physically crosslinked elastic networks.⁸⁵

In addition to stretchable electrodes, current collectors, and electrolytes, stretchable packaging, which is often neglected, has recently been studied to improve the long-term stability and safe operation of stretchable batteries. The liquid metals with both metallic and fluidic properties are demonstrated to achieve stretchable and hermetic sealing, providing a novel route to create stretchable battery devices.⁸⁶ Although significant progress has been made, the need for inactive elastomers, combined with limited loading of rigid active materials, often results in reduced volumetric energy density and electrochemical performance, resulting in a trade-off consideration between mechanical deformability and energy output.

Beyond intrinsically stretchable designs, kirigami architectures offer an alternative route to achieving mechanical deformability without fundamentally altering material compositions (Figure 5B). Kirigami designs utilize strategically placed mechanical cuts to impart stretchability to otherwise rigid battery films, enabling strains beyond the intrinsic limits of the materials themselves.^{58–60} These patterned cuts allow the battery to undergo complex deformations such as stretching, bending, and twisting. To demonstrate the potential of this design, Song et al. developed stretchable LIBs by cutting current collectors, electrodes, separators, and packaging layers into identical kirigami patterns before laminating them into a cohesive structure (Figure 7C),⁸⁷ achieving over 150% strain tolerance and stable cycling performance. Similarly, laser cutting has been used to fabricate composite electrodes on elastic substrates, allowing continuous

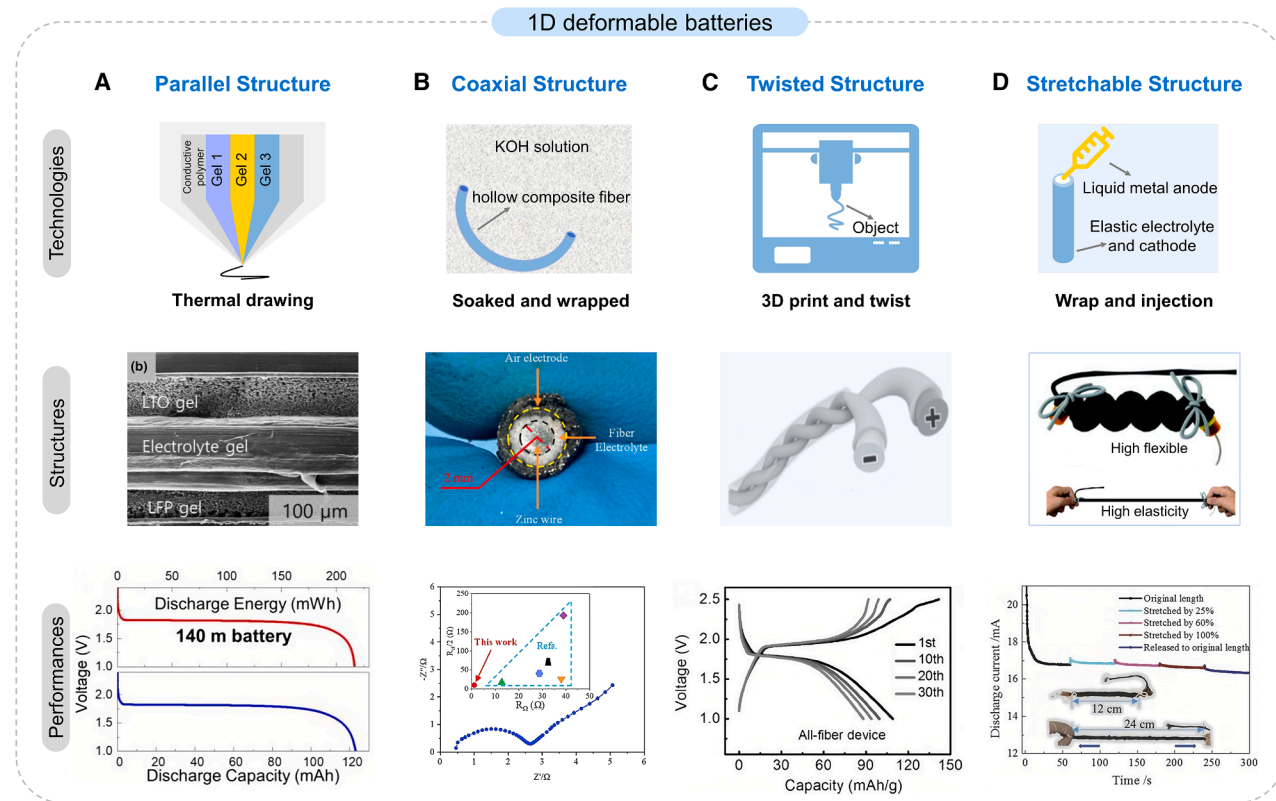


Figure 6. Designs and performances of 1D deformable batteries

(A) Parallel-structured long-fiber batteries prepared via thermal drawing: microstructure and charge-discharge performance. Reproduced, with permission, from Khudiyev et al.⁷⁰ Copyright 2022, Elsevier.

(B) Layer-by-layer wrapped coaxial fiber batteries: cross-sectional architecture and electrochemical impedance characteristics. Reproduced, with permission, from Zhang et al.⁴⁴ Copyright 2022, Elsevier.

(C) 3D-printed twisted fiber batteries: schematic structural diagram and corresponding capacity-voltage characteristics. Reproduced, with permission, from Wang et al.⁴⁶ Copyright 2017, Wiley-VCH GmbH.

(D) Stretchable fiber batteries fabricated via a wrapping method: optical images during stretching and discharge stability under different tensile strains. Reproduced, with permission, from Liu et al.⁵¹ Copyright 2018, Wiley-VCH GmbH.

deformation with minimal voltage loss.⁵⁸ Furthermore, by cutting flexible batteries into kirigami structures directly, batteries can deform simultaneously in folding, stretching, and twisting. For instance, a flexible ZIB using metal-coated polyurethane sponge electrodes and Ecoflex encapsulation could initially withstand only bending, but incorporating fence- or serpentine-like structures enabled stretchability up to 100% and 160%, respectively, while maintaining structural integrity and conductivity under repeated twisting.⁵⁹ These findings demonstrate that kirigami designs can effectively bridge the gap between mechanical adaptability and electrochemical performance. In particular, some patterned kirigami structures enable a 2D-to-3D transformation upon mechanical activation, making them promising candidates for future deformable batteries with dynamic response capabilities. Nonetheless, their practical implementation remains constrained by the geometry of the cuts, which dictates the deformation direction. Moreover, stress concentrations at sharp corners can cause mechanical failure, while the precision required for cut patterning poses challenges for large-scale manufacturing and long-term durability.

The bridge-island configuration represents another effective structural strategy for achieving deformability in planar batteries (Figure 7D).^{61,62} In this design, rigid battery segments (“islands”) are connected by flexible segments (“bridges”) composed of elastomers or conductive materials, enabling localized rigidity for energy storage while maintaining global mechanical flexibility. For example, in a LIB system, electrode islands laminated on a silicone elastomer substrate were interconnected using self-similar stretchable metal interconnects, resulting in excellent stretchability up to 300% and a stable areal capacity of approximately 1.1 mAh cm^{-2} .⁶² Similarly, in a ZIB, intrinsically stretchable liquid-metal “bridges” were screen-printed to connect active zinc-air battery (ZAB) islands, enabling seamless integration, high electrochemical stability, and customizable output through arbitrary series/parallel configurations.⁶¹ However, the fabrication process of bridge-island structure is relatively complex, involving multiple steps such as precise alignment and encapsulation. Moreover, increasing the coverage of active materials inevitably reduces the proportion of stretchable interconnects, which may compromise overall deformability. Further

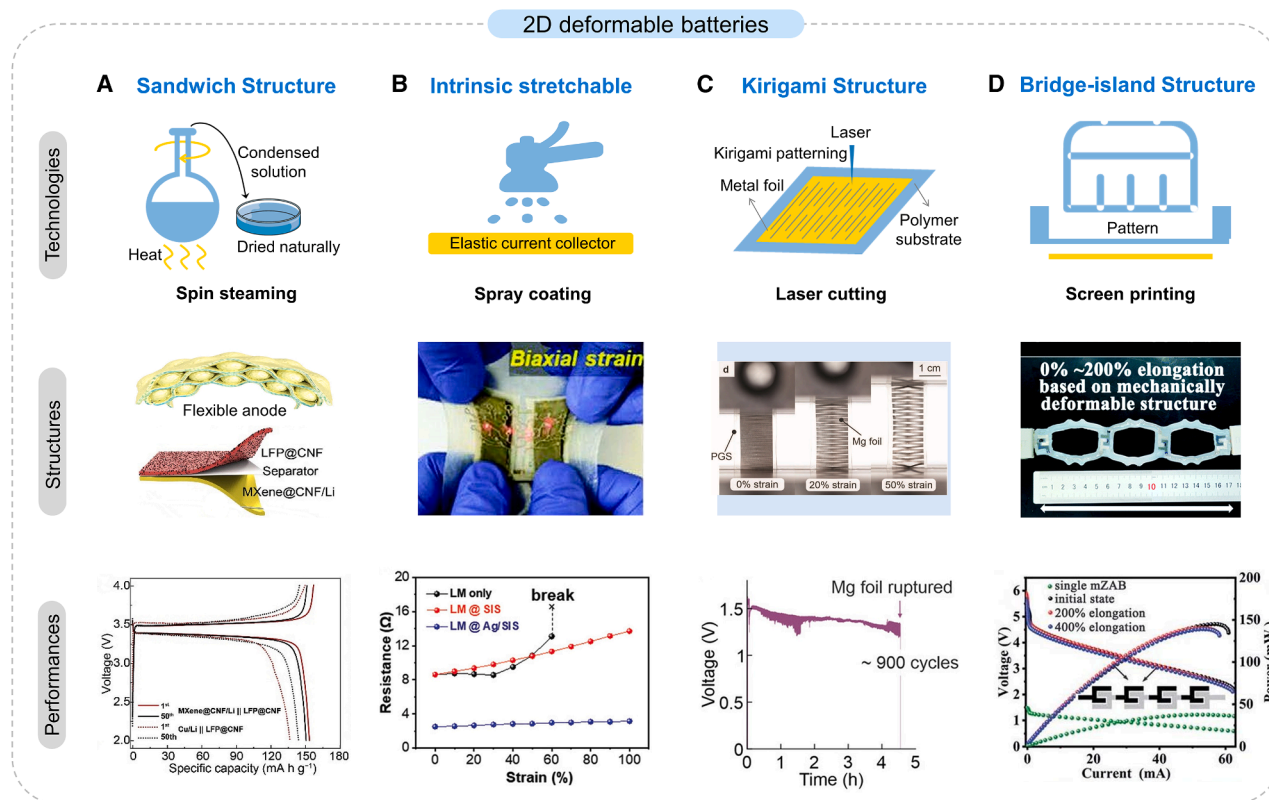


Figure 7. Designs and performances of 2D deformable batteries

(A) Sandwich-structured flexible battery fabricated via spin steaming: schematic illustration of electrode and battery assembly, and the charge-discharge voltage profiles. Reproduced, with permission, from Wang et al.⁶³ Copyright 2020, Elsevier.
 (B) Intrinsic flexible 2D battery prepared by spray coating: images of the stretchable circuit board directly printed on a stretchable battery, and the electrical resistance changes of circuit lines with strain. Reproduced, with permission, from Song et al.⁶⁴ Copyright 2020, Wiley-VCH GmbH.
 (C) Kirigami-structured flexible battery fabricated via laser cutting: photographs of Mg metal electrodes under different strains, and discharge of battery with 50 μ A when subjected to cyclic straining to 20%. Reproduced, with permission, from Karami-Mosammam et al.⁵⁸ Copyright 2022, Wiley-VCH GmbH.
 (D) Island-bridge structured flexible battery fabricated via screen printing: optical photograph of the flexible ZAB array at 200% elongation, and the discharge polarization curves under different deformations. Reproduced, with permission, from Zhao et al.⁶¹ Copyright 2021, The Royal Society of Chemistry.

research is therefore needed to optimize interconnector geometries and integration strategies to enhance both stretchability and energy performance.

Following the development of planar battery architectures, 3D design strategies have emerged to provide flexible mechanical functions and considerable design freedom by folding or bending planar batteries into spatial forms (Figure 5C). Wavy structures^{63,64} and origami^{65–67} models have been employed to realize deformable batteries (Figure 8).

As one representative 3D strategy, the wavy structure enables cell stretchability through geometrical transformation without altering the intrinsic mechanical properties of the materials (Figure 8A). Generally, foldable but non-stretchable material is first tightly affixed to a pre-stretched elastic substrate. After the pre-stretched substrate is released, its original shape is restored, and the material affixed to it spontaneously forms wavy wrinkles periodically. Based on this pre-stretch method along with electro-deposition, a flexible and stretchable ZAB with a wavy structure have been produced (Figure 8A, row 2).⁶³ Such a battery is stretched or compressed with a structural transformation from

2D to 3D with promising charge and discharge ability under different tensile strains (Figure 8A, row 3). Similarly, a stretchable NiCo-Zn battery was fabricated using this structure.⁶⁵ In the battery, NiCo@Au@CNT and Zn@Au@CNT paper electrodes are placed on each side of a stretched electrolyte, and sodium polyacrylate (PANA) hydrogel electrolyte is prepared as an elastic substrate to be pre-stretched. Eventually, over 400% tensile strain is achieved with no capacity wastage. In another work, a stretchable ZIB with a combination of woven and wavy structures was created. Metal Zn and Ni electrodes are supported via dynamic stretching-electroplating on a wavy commercial spandex textile covered by a thin Cu layer, and a stretchable polymer is utilized as the electrolyte. The battery can withstand 100% uniaxial strain and maintain stable electrochemical performance under repeated stretching-releasing deformation. However, due to the unidirectional stretching of the elastic substrate, the battery can only stretch in the direction of wave propagation, which limits its application scenarios.

The origami technique, involving the use of foldable materials and mechanical structural design, presents another promising

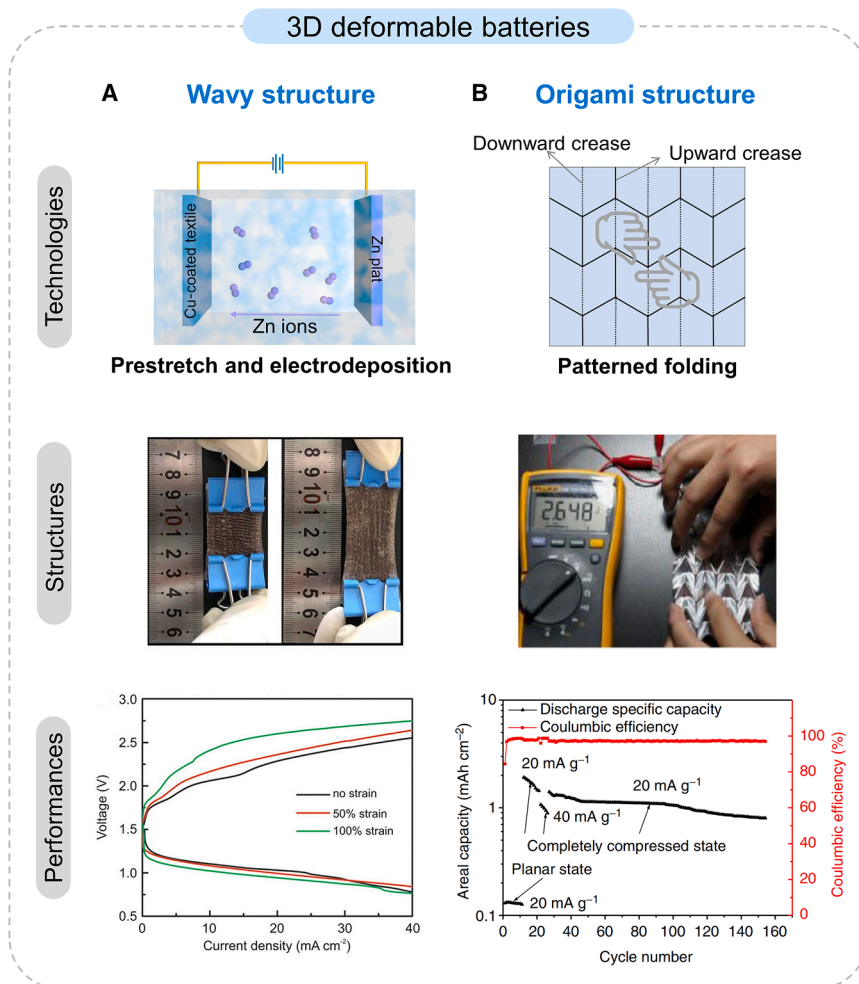


Figure 8. Designs and performances of 3D deformable batteries

(A) Wavy-structured 3D flexible ZAB fabricated via pre-stretching and electrodeposition: photographs of the Cu-coated textile substrate before and after stretching beyond 100%, and charge-discharge polarization curves of the ZABs under different strains. Reproduced, with permission, from Qu et al.⁶³ *Carbon Energy*, 2022, under creative commons attribution license (CC BY).

(B) 3D deformable battery formed by patterned folding: photograph of fold deformation of an origami LIB while connected to a voltmeter, and its capacity retention and Coulombic efficiency as a function of cycle number. Reproduced, with permission, from Song et al.⁶⁷ Copyright 2014, Springer Nature.

used simultaneously in a deformable LIB design.⁸⁸ Before the bending step, the processing technology and materials of an origami battery are the same as those used in conventional paper batteries, so various conventional modification and component integration methods apply to origami batteries. Similar to the flexibility and functionality of kirigami batteries, those of origami batteries are significantly constrained owing to the given patterns in battery designs. The origami technique is ineffective for small battery systems with high volumetric energy density due to the considerably higher dynamic space required by the 2D-to-3D conversion. Therefore, suitable application scenarios are yet to be developed

approach for constructing deformable objects without relying on stretchable materials (Figure 8B). The origami technology is reconfigurable and expandable, using the science of folding flat batteries following pre-defined folds to permit continuous and direct 2D-to-3D transformations. Various origami-based approaches have been pursued in practical applications, including solar panels, robotic arms, and heart stents. The origami method is also gradually being used in the design of deformable batteries. In an early work, a conventional planar LIB consisting of CNT-coated paper current collectors with deposited active material slurry and standard commercial packaging, a deformable origami LIB was created by folding in the Miura pattern (Figure 8B, row 2).⁶⁷ The obtained origami LIB achieves both linear and areal deformability, high twistability, and high bendability. Moreover, in the fully compressed state, the areal capacity remains above 1 mAh cm⁻² over 110 cycles, demonstrating reasonably good electrochemical performance (Figure 8B, row 3). However, increasing the amount of active material will decrease the rate performance and produce higher local strains at the creases, limiting the area capacity improvement of battery. Origami can also be used in conjunction with other design strategies. For example, island-bridge and origami structures were

for wavy and origami deformable batteries to foster strengths and circumvent weaknesses, such as applying origami batteries as robot and wing materials for simultaneous power supply and extension-contraction conversion.

Microbatteries

Microbatteries are suitable for electronic devices that require small batteries, such as microrobots, sensors, wearable devices, and medical devices (e.g., cardiac pacemakers and neurostimulators), and have been studied widely (Figure 9).⁸⁹ Short-fiber batteries^{90,91} are the simplest type of microbatteries, followed by 2D (flat) microbatteries,^{92,93} which have limited surface areas. To address the low energy density of miniature batteries, 3D miniature batteries with integrated structures and component materials are designed.^{94,95} Generally, batteries that meet the following conditions are classified as microbatteries: (1) 1D fiber batteries less than 3 cm in length; (2) 2D microbatteries with an area in the square-centimeter range and a thickness of less than 10 μm; and (3) 3D microbatteries with a volume of less than 10 mm³. The tight stacking of battery components is the key to miniaturization. With the limited battery size, component materials and structural design should be optimized to ensure

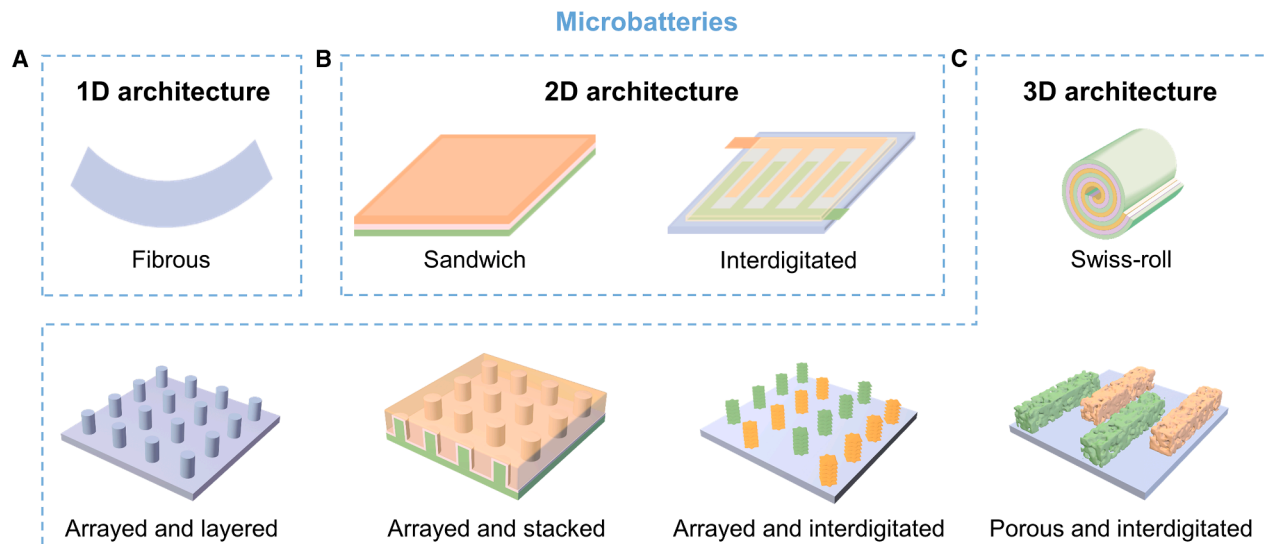


Figure 9. Structural engineering of microbatteries across different dimensions

(A) 1D architecture: fibrous.^{90,91}

(B) 2D architecture: sandwich¹⁰⁰ and interdigitated.¹⁰⁴

(C) 3D architecture: swiss-roll,¹¹³ arrayed and layered,¹⁰⁷ arrayed and stacked,¹⁰⁸ arrayed and interdigitated,¹⁰⁹ and porous and interdigitated.¹¹⁰

the adequacy of electrochemical properties. A complete description of the main structural design strategies to construct microbatteries in different dimensions is given below.

1D microbatteries are well suited for applications that demand mechanical flexibility and linear integration, such as smart textiles and wearable electronics (Figure 9A). However, due to their small package area and low percentage of active materials, fibrous 1D microbatteries often exhibit limited electrochemical performance. Therefore, enhancing energy density has become a primary objective in their structural design.⁹⁶ Two main strategies have been explored to address this issue: advanced electrode material engineering and the construction of compact battery architectures. One example of electrode material engineering has been demonstrated in a high-performance Zn dual-ion fiber microbattery (Figure 10A).⁹⁷ In this system, the anode and cathode are arranged in a parallel architecture, where the contact area between the two electrodes is inherently limited. To overcome this limitation, both electrodes were engineered to enhance interfacial kinetics. The Zn fiber anode was coated with a nanometer-thick conductive carbon layer via sputtering, which suppressed dendrite formation and stabilized the electrode-electrolyte interface. Meanwhile, the cathode employed a core-shell structure consisting of a high-conductivity fiber core and a polyaniline (PANI) sheath. This design enabled the Zn-ion microbattery to deliver high energy density and remarkable rate performance. Another strategy focuses on minimizing the device footprint through compact integration. A representative example is a tube-in-tube microbattery design (Figure 10B),⁹⁸ where a metal-coated patterned membrane was rolled into a microtube electrode and integrated with a Ti wire counter electrode inside a glass capillary. This compact configuration, with an outside diameter of 0.7 mm and a footprint of 0.39 mm², delivers a high areal energy density of 313 μWh

cm⁻² and a power density of 52 μW cm⁻², outperforming conventional micro-LIBs. However, 3D microtubular electrodes formed by the self-assembly of 2D nanomembranes have structural defects. The potential of fiber microbatteries can be fully realized by remotely controlling the behavior of 2D membranes during assembly while under the influence of external magnetic fields.⁹⁹

The limitations of 1D microbatteries in electrochemical performance have motivated the development of 2D microbatteries, which offer advantages in structural uniformity, areal utilization, and integration compatibility. There are two fundamental structural types for 2D microbatteries, as shown in Figure 9B: the layer-by-layer stacking sandwich structure and interdigitated structure. The layered stacking structure has a proven processing method and does not require intricate patterned battery components.¹⁰⁰ However, under a fixed footprint, a high energy density requires thick electrodes with an increased ion-transport distance, which poses a challenge to battery power performance.¹⁰¹ Microbatteries with interdigitated structures have alternating cathodes and anodes placed at certain intervals on the substrates, and thus the ion-transport distance and battery power performance can be controlled.¹⁰² However, the presence of isolation bands significantly reduces area utilization, resulting in a low energy density for a given battery thickness. While developing high-energy-density electrode materials and high-ionic-conductivity electrolytes, researchers have also tended to reduce the use of common inactive materials (e.g., packaging, binders, and conductive agents) in order to improve the electrochemical performance of planar microbatteries. A small (20–40 μL) and light (50–80 mg) microbattery in sandwich structure is made possible by combining the advances in battery architecture, chemistry, and packaging (Figure 11A).¹⁰³ The metal current collectors also serve as hermetic shells for

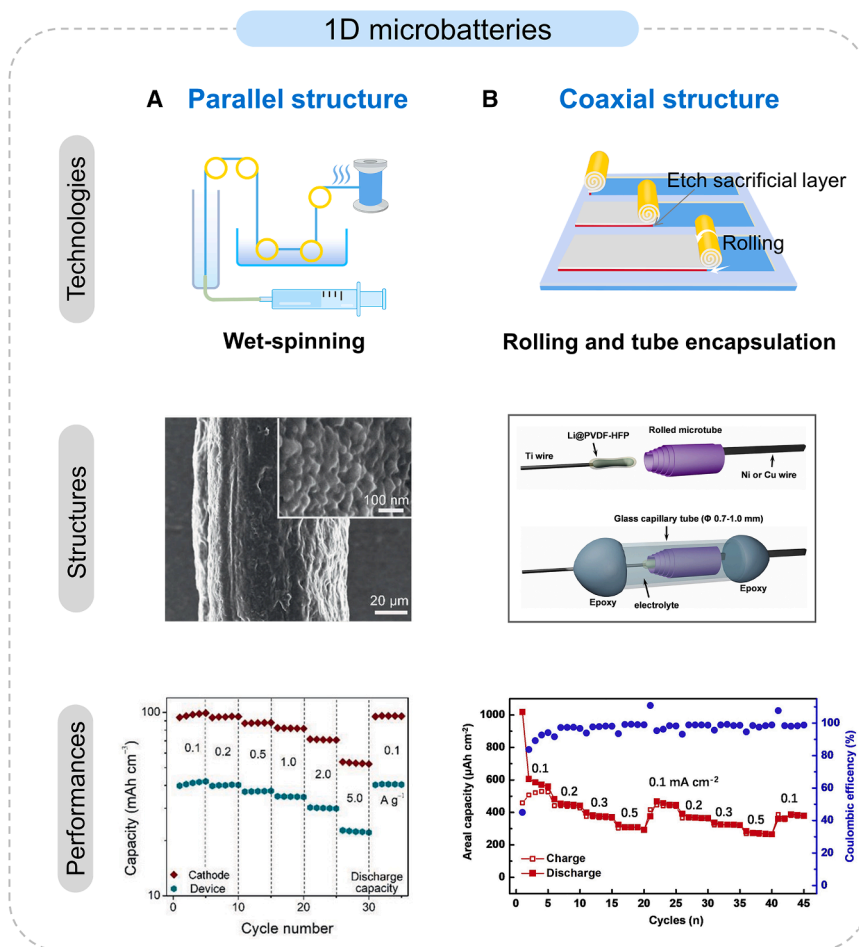


Figure 10. Designs and performances of 1D microbatteries

(A) Parallel fiber microbattery fabricated via wet-spinning and electro-polymerization: SEM image of the cathode fiber and volumetric capacity at different current densities. Reproduced, with permission, from Zhai et al.⁹⁷ Copyright 2021, Wiley-VCH GmbH.

(B) Coaxial fiber microbattery fabricated via rolling and tube encapsulation: schematic illustration of the structure of tube-in-tube micro-LIB and the rate capability. Reproduced, with permission, from Weng et al.⁹⁸ *Cell Reports Physical Science*, 2021, under CC BY-NC-ND license.

the microbatteries, which greatly reduce the mass and volume of the packaging compared to macroscopic battery packaging designs where electrodes are assembled into an external hermetic shell. In the planar interdigital structure, an *in situ* electrodeposition strategy is adopted to construct a flexible aqueous Zn-I microbattery with unprecedented energy density.¹⁰⁴ The interdigitated microelectrodes are firstly manufactured by combining high-resolution laser direct writing and the customized stainless-steel template, after which *in situ* electrodeposition of iodide occurs during the charging process without binder and conductive agents (Figure 11B).¹⁰⁴ The high-ionic-conductivity polyelectrolyte ensures that the as-fabricated microbattery has a high volumetric and areal energy density, and the appropriate micro-electrode arrangement can effectively inhibit the shuttling effect of I^{3-} .

The construction of 3D miniature batteries can elevate the energy density and power performance needed from miniature batteries, providing more structural design space. To this end, some researchers fabricated electrodes into 3D structures, mainly by introducing random/regular porous structures and constructing nanoarray structures.^{105,106} A 3D electrode can be created by the structural design of the active material itself, or the active material can be cladded on a manufactured 3D cur-

rent collector. As in 2D microbatteries, positive and negative electrodes can be assembled in layered or interdigitated structures. Five typical 3D microbattery structures reported in the literature are depicted in Figure 9C: (1) arrayed and layered, where cathode and anode are coated on microarrays layer by layer; (2) arrayed and stacked, where one electrode is closely filled in the gap of another electrode microarray, and the electrolyte layer separates the two electrodes; (3) arrayed and interdigitated, where cathode columns and anode columns are interleaved on the substrate; (4) porous and interdigitated, where porous cathodes and porous anodes are alternately arranged on the substrate; and (5) Swiss roll, where the flat battery films are assembled into a Swiss roll. Compared to 2D microbatteries, 3D microbatteries generally exhibit better power performance, and a higher degree of compactness of the active material is more conducive to energy density enhancement. However, since specific processing methods such as etching and sputtering are usually required in the manufacture of 3D structures, the processability will be reduced to some extent. For instance, a microbattery was fabricated based on the arrayed and layered structure (Figure 12A). The etching of the Si wafer is applied to ensure the robust scaffold, and an insulating layer, a current collector, a negative electrode, and the solid electrolyte are step-conformally deposited on the 3D Si microtube pattern via atom layer deposition (ALD).¹⁰⁷ The arrayed structure provides high area enlargement and enhances the material mass loading while keeping a small footprint area. Combined with the thick deposited layers, the area specific capacity can be increased compared to the planar counterpart. Similarly, filling the gaps of the scaffold completely with electrode material instead of just an electrode material layer cladding can effectively improve the active load, opening a promising route for fabrication of powerful and energy-dense microbatteries (Figure 12B). In one case, to form this compact structure, the crystalline Si wafers are etched as anode, which is a reliable and common place process.¹⁰⁸ Notably, the arrayed

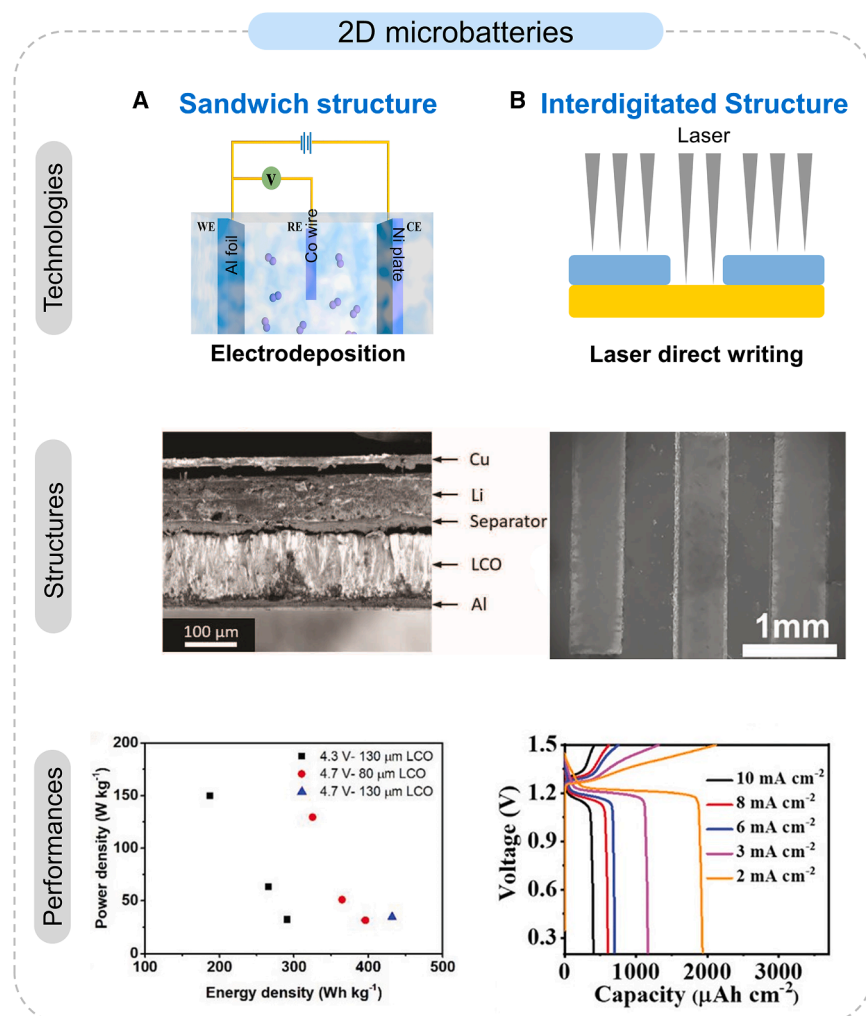


Figure 11. Designs and performances of 2D microbatteries

(A) Sandwich-structured microbattery fabricated via electrodeposition: SEM cross-sectional image and Ragone plot of fully packaged microbatteries. Reproduced, with permission, from Yue et al.¹⁰³ Copyright 2021, Wiley-VCH GmbH.

(B) Interdigitated microbattery fabricated via laser direct writing: SEM image of a single microelectrode and galvanostatic charge-discharge curves of the zinc-iodine microbattery at various current densities. Reproduced, with permission, from Jin et al.¹⁰⁴ Copyright 2022, Wiley-VCH GmbH.

strategy significantly reduces the footprint of microbatteries and improves compatibility with mass production and microelectronic circuitry. Qu et al. made the on-chip Swiss-roll current collector via this method to hold MnO₂ electrode slurry.¹¹³ The millimeter-scale micro-ZIB using the MnO₂ Swiss-roll microcathode and Zn anode is small enough to be placed on the edge of a coin, and a digital timer with low power consumption can be powered by this micro-ZIB for more than 5 h.

The 3D miniature batteries exhibit excellent and balanced electrochemical performance. Nevertheless, manufacturing 3D substrates and achieving precise, uniform, and strongly adherent conformal deposition of active materials within holes or channels remain challenging. Furthermore, it is better for deposition processes to have good compatibility with existing microfabrication processes.

and stacked structural design presents a long ion-diffusion path for accessing the entire capacity of the cathode; thus, electronic and ionic conductivity must be considered and maximized.

Based on interdigitated electrodes, 3D microbatteries can be fabricated in arrayed-interdigitated and porous-interdigitated configurations, as shown in Figures 12C and 12D.^{109,110} For example, Li et al. used a porous substrate to form the porous and interdigitated structure of a high-performance 3D potassium-ion microbattery.¹¹⁰ In a separate study, the cathode and anode are integrated on two polarities of interdigitated 3D nanoporous Au current microcollectors using femtosecond laser scribing and subsequent multistep electrodeposition.¹¹¹ These 3D electrodes enhance electrical conductivity and wettability, ensure intimate contact with active materials, and shorten the ion-diffusion distance between cathode and anode. In recent years, the 3D Swiss-roll architecture (Figure 12E) microbatteries have been developed by borrowing from the manufacture of energy-dense cylinder batteries. The 2D flat layer stack can be self-assembled into rolled-up microtubes when the layers are delaminated and the built-in strain is released.¹¹² This structural

Meanwhile, the manufacturing efficiency of the proposed innovative processing method needs further study.

All-in-one batteries

In traditional batteries with sandwich (stacked) configuration (Figure 13A), the adjacent components (such as the electrode, separator, and electrolyte) rely on externally applied pressure to maintain point-to-point contact. When such a battery is exposed to mechanical deformation or insufficient external pressure, dislocation and detachment occur between the neighboring component layers, resulting in unstable interfaces, reduced contact area, and, ultimately, failure of electron/ion transport that terminates battery operation.¹¹⁴

In contrast, all-in-one batteries integrate all functional components (including current collectors, electrodes, electrolyte, and encapsulation) into a unified monolithic architecture (Figure 13A). This design eliminates the need for pressure-dependent assembly and promotes strong interface affinity and efficient electron/ion transport, enabling superior stability under mechanical stress and during fast charging/discharging. Moreover, the compactness of the structure and reduced dosage of inactive materials, such as

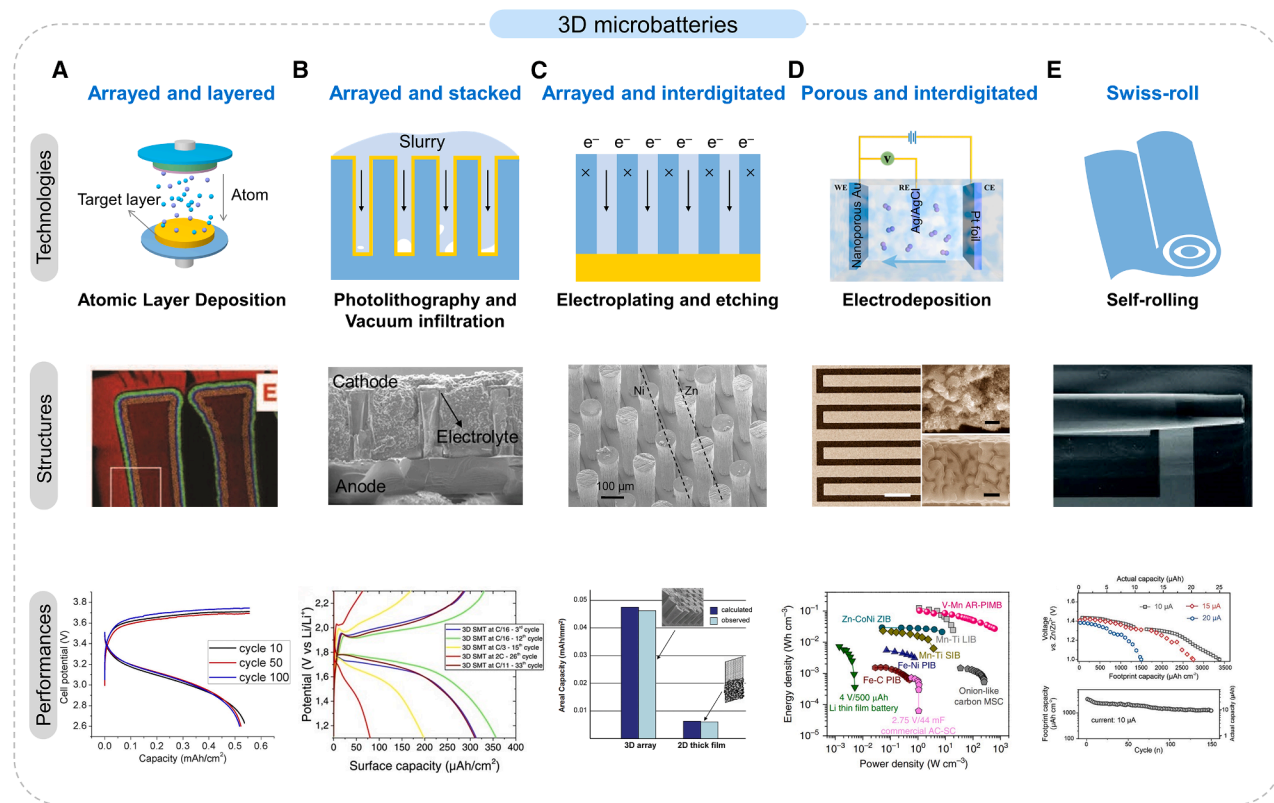


Figure 12. Designs and performances of 3D microbatteries

(A) Microbattery with arrayed and layered architecture: schematic illustration of the main ALD fabrication technique, energy dispersive X-ray spectroscopy-transmission electron microscope mapping of the coated array, and the charge-discharge surface capacity evolution of the micro-LiB. Reproduced, with permission, from Letiche et al.¹⁰⁷ Copyright 2017, Wiley-VCH GmbH.

(B) Arrayed and stacked architecture microbattery fabricated via photolithography and vacuum infusion: SEM image of the full 3D battery and galvanostatic cycling curves of the porous 3D cathode array. Reproduced, with permission, from Hur et al.¹⁰⁸ Copyright 2018, Elsevier.

(C) Arrayed and interdigitated microbattery fabricated via electrodeposition and etching: SEM image showing interdigitated Ni and Zn electrodes, and capacity comparison of the mesocarbon microbead electrodes with 3D arrays and conventional 2D thick film configurations. Reproduced, with permission, from Chamran et al.¹⁰⁹ Copyright 2007, IEEE.

(D) Porous and interdigitated architecture microbattery fabricated via electrodeposition: SEM image of interdigital-patterned nanoporous Au current collectors, anode and cathode, and the stack energy and power densities of the fabricated potassium-ion microbattery. Reproduced, with permission, from Li et al.¹¹⁰ *Nature Communications*, 2019, under a creative commons attribution 4.0 international license.

(E) Swiss-roll structured microbattery fabricated via self-rolling: SEM image of a bare Swiss-roll microelectrode, and galvanostatic discharge curves and cycling performance. Reproduced, with permission, from Jin et al.¹¹² Copyright 2019, Wiley-VCH GmbH.

binders and current collectors, help in decreasing the weight and thickness of bulk batteries; such batteries are ideal for application scenarios that require wearability, endurance, and fast charging.¹¹⁵ However, it is difficult to build seamless interfaces due to differences in the preparation conditions and methods of battery components and variations in the physical states and chemical properties of battery materials. Unique processing and adjacent-component-layer selection can effectively help to construct structurally designed batteries. Here, we summarize the three main strategies of fabricating high-performance all-in-one batteries (Figure 13B).

Liquid penetration

Using liquid penetration of the battery component slurry to enhance interfacial continuity is the most common method for building all-in-one batteries. As shown in preparation strategy I in Figure 13B, the slurry spreads across the substrate and, due

to its fluidity, can penetrate into the microchannels or pores of the rough surface under applied pressure or sufficient standing time. During the subsequent drying process, the slurry transitions into a solid or quasi-solid state, resulting in strong physical interlocking and interfacial adhesion. This transition ensures structural stability and mechanical robustness, ultimately leading to the formation of a seamless all-in-one battery.

A representative implementation of the liquid penetration strategy is the blade-coating method, which is commonly used to fabricate film-based components such as active material layers, separators, and solid electrolytes. This method has the potential to eliminate the need for an extra current collector and is easy to scale up, although it is generally difficult to apply with commercial separators. In this process, a viscous slurry containing target components is spread onto the substrate (e.g., glass fiber separator and polyimide membrane) under

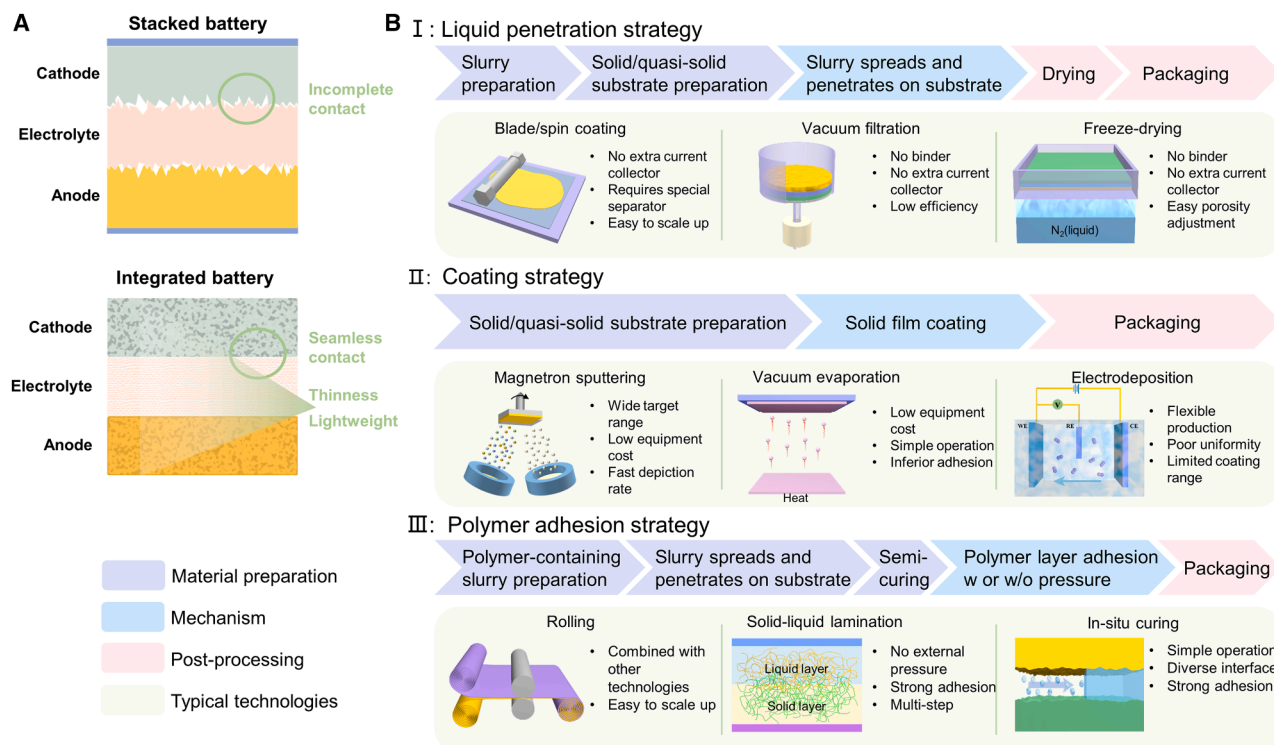


Figure 13. Structural characteristics and processing methods of all-in-one batteries

(A) Interfacial characteristics in traditional and all-in-one batteries. Traditional batteries have incomplete and poor interfacial contact, whereas all-in-one batteries deliver seamless, tight interfacial contact.

(B) Typical manufacturing processes for all-in-one batteries.

shear stress and fills the grooves under compressive force to form a composite film.¹¹⁶ Similar to the blade-coating method, spin coating was used to create integrated sodium-ion batteries (SIBs), in which the cathode and anode slurries are cast on both sides of a modified polyethylene separator to generate intimate interfacial contact.¹¹⁷

Vacuum filtration is widely used to prepare gradient materials by controlling the liquid composition and filtering sequence. Similarly, layered structurally designed batteries can be produced by dispersing the electrode and separator materials in solutions and then carefully sifting them. An aqueous ZIB was constructed by three-step vacuum filtration.¹¹⁸ After freeze drying and pressing, the obtained monolithic filter cake can form a composite film and turn into an all-in-one ZIB by immersion in the electrolyte for a certain time. Due to the penetration of the upper layer solution, the interfacial contact between the separator and electrodes is robust and complete. This method eliminates the need for binders and extra current collectors and results in thin, compact batteries with excellent electrochemical performance, even under deformation. However, the overall fabrication efficiency remains relatively low, limiting its scalability.

In addition to water content reduction, freezing can facilitate transition from the liquid state to the solid/quasi-solid state, so it is feasible for structurally designed batteries. The freezing order of suspensions determines the integrated structure, whereas the voltage and concentration of suspensions can regulate the

amount, thickness, and porosity of the active material layers and the separator layer. With use of this technique, a Teflon mold is placed on liquid N₂, and suspensions of graphene-polyaniline (GO-PANI), cellulose nanofiber (CNF), and exfoliated graphene (EG)-Zn are successively poured into the mold after the first suspension completely freezes into a solid.¹¹⁹ The resulting ice monolith is freeze dried and pressed to form an integrated film, and all-in-one ZIBs are finally assembled by adding electrolyte.

Liquid penetration requires the battery components to be able to form a homogenous liquid in an appropriate solvent, since the preparation process must begin in the liquid stage. Freeze drying and vacuum filtration are challenging for solid battery components (e.g., Zn foils and glass fiber separators), which are difficult to dissolve into solutions.

Coating

Physical or chemical coating of the substrate surface enables the formation of thin films with good adhesion and is widely used in the structural design of batteries, as shown in Figure 13BII. In integrated battery manufacture, physical vapor deposition (PVD) and electrochemical deposition are common coating processes. In PVD, functional films are deposited on substrate surfaces through the evaporation and migration (e.g., magnetron sputtering, ion sputtering, and vacuum evaporation) of material sources. Magnetron sputtering features a wide target range, low equipment cost, and fast deposition rate, while vacuum evaporation

offers simple operation but suffers from poor adhesion.¹²⁰ The electrodeposition method of coating is simple and flexible in operation and does not require complex equipment, but the formed coating also has poor adherence and may produce polluting effluent.¹²¹ Since a single coating layer can only be constructed in one step, integrated battery manufacture requires multistep operations or combinations with other integrated battery manufacturing processes to produce a wholly integrated battery.

Polymer adhesion

The polymer adhesion strategy plays a vital role in the integration of battery components, particularly in systems that employ polymer-based interface layers or polymer electrolytes, as illustrated in Figure 13BIII. Typically, polymer-containing slurries are spread over the substrate and partially cured to form shaped components. Adjacent components then adhere under applied pressure or heat, forming continuous interfaces. This approach enhances interfacial contact and mechanical robustness without the need for external binders or adhesives. The effectiveness of polymer adhesion stems from the high motility of chain segments in adjacent polymer layers, especially in solution or under external stimuli such as heat, radiation, pressure, and pH. Under such conditions, the segments readily form physical (e.g., ionic and hydrogen bonds) and chemical crosslinks, significantly improving interfacial adhesion. Common processing methods include rolling, layer-by-layer polymerization, and *in situ* curing. Rolling can be combined with other technologies and is easy to scale up, making it suitable for large-scale integration. In contrast, layer-by-layer polymerization and *in situ* curing do not require external pressure and offer simple processing with broad interfacial compatibility. For example, polyvinyl alcohol (PVA)-Zn(CF₃SO₃)₂ gel electrolyte layers are doctor bladed on the anode and cathode films of a ZIB.¹²² During the heating and pressing-rolling, a large number of hydrogen bonds are constructed between the opposing polymer layers, and the PVA chain segments become entangled, finally forming the integrated ZIB through the fusion of electrolyte layers. Without external force, polymer layers can undergo physical-chemical crosslinking at certain temperatures.¹²³ Thus, the fabrication of all-in-one batteries via solid-solid crosslinking is easily scaled up. Since bonding between polymer layers is required, this method is not applicable to batteries that have liquid electrolytes. Moreover, there is often no polymer layer on the electrode surface, so an additional decorated polymer film will increase the mass of inactive materials in a battery, thus contradicting the need for all-in-one batteries in lightweight scenarios.

Integrated electrodes (e.g., integration of current collectors and cathodes/anodes) are drawing considerable attention, but few studies cover full all-in-one batteries. This is because the adhesion of multiple interfaces is exponentially more difficult than that of single interfaces. New manufacturing techniques, such as 3D printing, are being used to manufacture all-in-one batteries. These approaches are expected to address the challenges in processing all-in-one batteries with multiple steps and complex operations.

Structural batteries

The development of high-energy-density battery systems, epitomized by LIBs, has progressively increased the electric power

supply in electric vehicles, aerospace, and marine applications, among others. However, when batteries are integrated into power equipment, additional mechanical protection components are often required to mitigate the impacts of applied loads on the battery, resulting in increased size and reduced overall energy density. Structural batteries combine energy storage and mechanical load-bearing functions, enabling the reduction in weight and size of the power system, which improves the systemic energy efficiency and reduces waste materials and carbon emissions.

The design of structural batteries can be classified into two main categories: coupled and decoupled structural batteries (Figure 14A). This classification is based on the integration of components responsible for electrochemical storage and mechanical load bearing within the structural battery system. In coupled structural batteries, multifunctional materials are employed as components capable of withstanding mechanical stresses while facilitating electrochemical reactions. In contrast, uncoupled structural batteries comprise mutually independent single-function components, with each element responsible for a distinct electrochemical or load-bearing role. This section will focus on the material selection and structural composite design of unit structural batteries, while modular and large-size structural batteries, which are built based on the integration of multiple cells, will be discussed subsequently.

Multifunctional composites represent a crucial element in the development of coupled structural batteries. To achieve the desired mechanical strength and stable performance of the entire battery system, it is essential to design all the battery components simultaneously, including electrodes, current collectors, electrolytes, separators, and auxiliary materials (Figure 14B). High-strength multifunctional materials with sufficient ion/electronic conductivity can be employed as structural battery components directly, such as CF, metal foil, and solid electrolytes.²⁵ Concurrently, a considerable number of battery materials that are not mechanically strong must be compounded with high-mechanical-strength materials to achieve versatility.

For the structural electrodes or current collectors, metallic materials with diverse morphologies (including sheet, net, foam, and scaffold structures), often possessing high strength (as observed in materials such as Al, Cu, Ti, and stainless steel) and outstanding electrical conductivity, have been widely employed as structural collectors. In the earlier work of Eckstedt et al., the Cu mesh collector and Al fiber weave collector were composited with CF weave and lithium iron phosphate (LFP) as anodes and cathodes, respectively, to validate a prototype of fiber-reinforced structural batteries.¹²⁴ In the case of softer metals (e.g., alkali metals, Ag, and Au), they are employed as structural conductors/anodes in the form of thin coatings to enhance conductivity or composites with reinforced matrix.¹²⁵ Additionally, CFs serve as core materials for current collectors, offering the potential to reduce non-active mass due to their high mechanical stability, lightweight nature, and electrical conductivity. The two most utilized continuous CFs are pitch-based and polyacrylonitrile-based, while the former is highly graphitized and the latter exhibits a disordered structure that is conducive to ion intercalation.¹²⁶ Using CF as both the current collector and reinforced structural material, the Young's modulus of the Zn-MnO₂

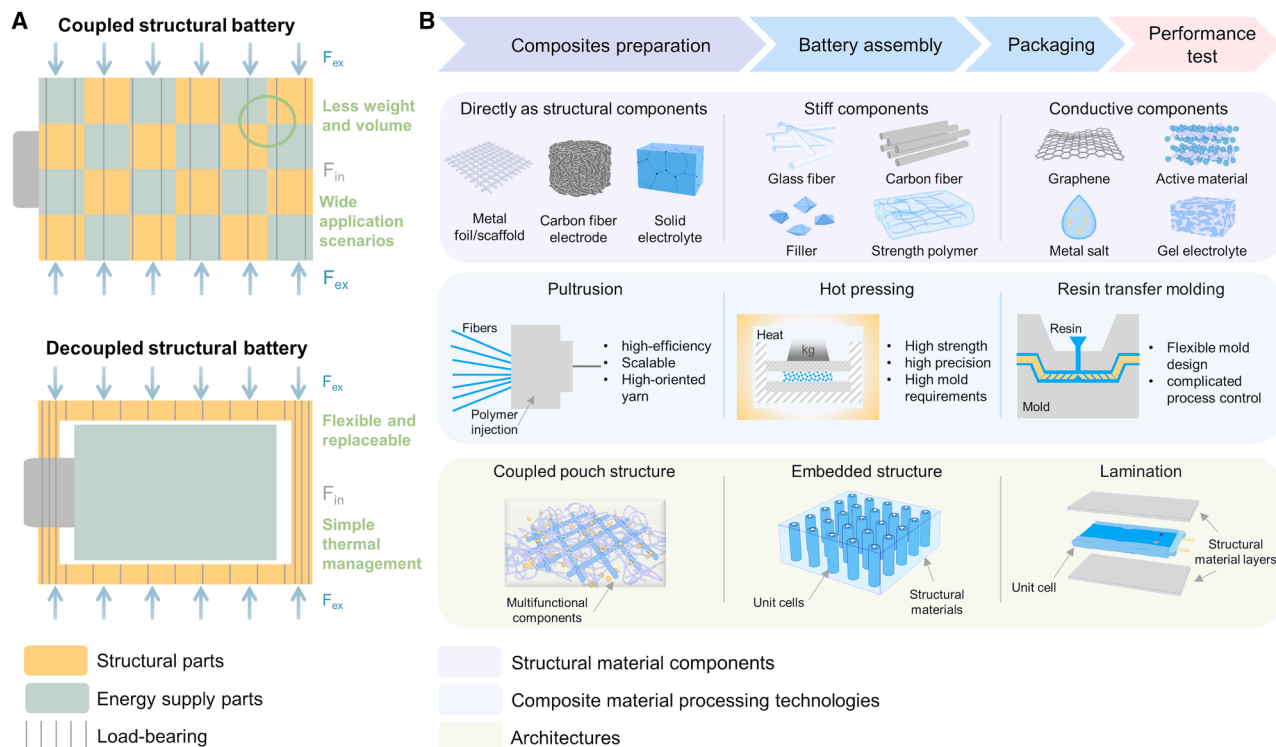


Figure 14. Characteristics and processing methods of structural batteries

(A) The integration state of structural parts and energy supply parts in coupled and decoupled structural batteries.

(B) Composite materials, processing technologies, and architectures for structural battery production.

structural battery is as high as 12.8 GPa, and the ultimate tensile strength is 293.4 MPa, which is higher than that of automotive aluminum alloy materials. Benefiting from the improved mechanical stability, the battery has a capacity retention rate of 88% after 100 cycles.¹²⁷ Meanwhile, CFs when employed as structural active anodic materials also show numerous effective results and potential. The CF was utilized as both structural anode and current collector loading LFP cathode materials, which were subsequently embedded in a biphasic solid-liquid structural battery electrolyte to construct an all-CF-based structural battery.²¹ The overall CF volume fraction is 23%, and the structural LIB has an excellent elastic modulus of more than 76 GPa in the fiber direction, which is the highest value reported in the literature so far, indicating its potential for replacement in electric vehicle structural parts. However, commercial CFs possess low reversible capacity ($\sim 230 \text{ mAh g}^{-1}$), and the improvement is still limited even after surface modifications such as acid or heat treatment to regulate surface functional groups or microstructures.¹²⁸ Therefore, many studies report compounding CFs with a high-specific-capacity anode (e.g., Sn, SnO_2 , Si, or Co_3O_4) as a direct and effective method to address this issue while maintaining favorable conductivity and mechanical strength. To enhance the inter-connectivity between CFs and traditional battery materials and to guarantee a unified and unwavering interface, the integration of structural adhesives is typically employed. Polymer materials are the mainstay of battery binders, as they have strong adhesion and self-healing proper-

ties and can be combined with a wide range of substrates. The most basic application is to prepare elastic electrodes by mixing elastic binder with electrode material. Furthermore, by designing the interface structure of binder-substrate, the binder adhesion can be enhanced. Inspired by the deep and strong root network structure of trees, the structure of polymeric binding materials was designed to hinder sliding between different layers inside a cell and increase cell flexibility. The binders were infiltrated into the porous cathode and anode to form a continuous network and then laminated to a ceramic-coated separator through hot pressing. This tree-root-like structure of the binder network enhances the cell's mechanical stability as well as the bending modulus by 11-fold to 3.1 GPa at appreciable deflections. Benefiting from the small thickness and limited redundancy of the adhesion layer, the specific energy of the cell is only reduced by 3%. Meanwhile, high-molecular-weight binders are beneficial to dense and robust battery structure. Furthermore, fiber-reinforced polymer materials have demonstrated utility as collectors and electrodes, exhibiting minimal degradation in mechanical properties following electrolyte wetting.¹²⁹

For structural electrolyte composites and separators, the mechanical reinforcement components (e.g., glass fiber, CFs, high-strength polymers, and inorganic fillers) are compounded with ionic conductors (e.g., solvent-containing/solvent-free electrolyte salts and ion-conducting polymer electrolytes) to form a homogeneous structure that ensures adequate and stable ionic transport. A novel non-fibrous highly amorphous cellulose

polymer electrolyte was developed for aqueous structural Zn-based batteries.¹³⁰ The exceptional mechanical robustness and electrochemical performance of Cellyte is achieved by embedding the structural cathode material onto CF substrates using Cellyte as a binder, leading to a continuous polymer electrolyte-cathode interface. The resulting structural battery demonstrates high electrochemical performance with excellent cycling stability over 1,200 h and more than 91.5% capacity retention. Additionally, the prepared structural battery is demonstrated to be highly environmentally adaptive, deformation tolerant, and abuse resistant, constituting a promising candidate for future low-cost, sustainable, and high-performance structural energy-storage designs. Furthermore, bicontinuous electrolytes with one phase providing high mechanical stiffness and another providing high ionic conductivity have shown promising performance.¹³¹ A bicontinuous electrolyte consists of a crosslinked bisphenol A ethoxylated dimethacrylate resin and Li bis(trifluoromethanesulfonyl)imide/fluoroethylene carbonate (LiTFSI/FEC)-diglyme-based liquid electrolyte was reported via polymerization-induced phase separation.¹³² By adjusting the liquid/solid ratio, the electrolyte can achieve an ionic conductivity of $1.62 \times 10^{-4} \text{ S cm}^{-1}$ and a modulus of 0.07 GPa at -10°C for a 90 wt % liquid/10 wt % solid composition, demonstrating the reference significance for the development of bicontinuous electrolyte low-temperature applications.

Given the stringent safety requirements for multifunctional batteries, material selection is critical for effective thermal management. Specifically, the thermal decomposition temperature of the electrolyte is a key parameter, and the selection of a highly thermally stable electrolyte can effectively slow down the thermal runaway process. Furthermore, separators, characterized by intrinsically low thermal conductivity, represent a thermal bottleneck. The application of ceramic-coated separators notably enhances thermal resistance, offering a significant strategy for improved thermal control. Additionally, as demonstrated by Liu et al.,¹³³ leveraging the latent heat absorption capacity of phase-change materials (solid-liquid or solid-solid) effectively maintains temperature uniformity and prevents overheating, thereby suppressing internal heat accumulation and enhancing the safety of LIBs.

In the decoupled approach, the structured battery system consists of a functionally independent battery part and a mechanical load-bearing part, which are integrated generally by embedded structure or laminate structure, while each part does not require multifunctional materials. Due to the low interaction between the cells and outermost surfaces, system-level electrochemical performance and functional properties are easier to design and balance.¹³⁴ Recently, an efficient 3D-printing-assisted fabrication strategy was proposed, whereby a distributed arrangement of internal batteries can be formed.¹³⁵ Sealed with lightweight CF reinforced composite plates, the structural batteries show a bending modulus of 24.5 GPa and retain an ultra-high capacity of 98.7% under a tensile stress of 80 MPa. In this circumstance, encapsulation and integration technologies and special processing conditions (e.g., heating, UV illumination, and water-oxygen content) are more challenging for the stability of cell performance and reinforcement materials. Due to the broad applicability to various functional unit cells,

encapsulation and integration techniques will be detailed in the next section.

Structural batteries show considerable potential for enhancing the power reserve of large electric devices and improving power supply stability under extreme usage scenarios. The development of high-performance multifunctional materials and the optimization of processing and integration technologies still offer significant opportunities for further advancement.

PACKAGING AND INTEGRATION FOR FUNCTIONAL BATTERY APPLICATION

Based on the development of functional materials and unit cells, integrated structural batteries will further contribute to the lightweight, high energy/power density, durability, and safety of large-format power systems.

Cylindrical cells are the most industrialized battery system and the main component of large-scale power packs (Figure 15A). They have been a major component of power engineering since the commercialization of LIBs. However, the application of cylindrical cells is limited to battery pack modules with specific sizes. The commercial cylindrical cell of 18650, by Sony Corporation, is 18 mm in diameter and 65 mm in height. Due to the standardization and consistency of 18650 cells on the market, it is considered the best candidate for battery pack construction.¹³⁶ The cylindrical batteries shown in Figure 15B are now being evaluated in industry for size, and the feasibility of different sizes in real devices has been verified. In recent years, Tesla has proposed a 4680 battery with a “tab-less” design that has a diameter of 46 mm and a height of 80 mm. This battery has five times the energy and six times the power of conventional commercial batteries.¹³⁷ A performance comparison between these batteries of different sizes suggests that reasonable battery sizes should be reconsidered for practical applications in different scenarios.

The volume cell-to-pack ratio and gravimetric cell-to-pack ratio of prismatic cells, a distinct commercial battery shape, can be enhanced to approximately 60% and 80%, respectively. This improvement is attributed to the innovative design of blade-shaped cells and the development of a modulus-free battery pack.¹³⁸ In addition, Tesla intends to use rigid integrated structural batteries as vehicle framework, as shown in Figure 15C.¹³⁹ This strategy will allow for a more rational allocation of space throughout the vehicle's energy supply system, similar to a wing structure for storing fuel.¹⁴⁰ It also requires that mechanically designed batteries for vehicles have extremely high rigidity and energy density throughout the special package.

Articular motion serves as a fundamental mechanism in human mobility. Nevertheless, achieving bending and other functionalities at each joint position requires substantial space. Hence, leveraging certain non-active spatial configurations can effectively balance the trade-off between space and energy utilization. Consequently, a flexible-joint battery is improved through imitation of the articular surface-ligament structures of biological joints to endow cylindrical structurally designed batteries with the flexibility of biological joints (Figure 15D) for adopting various positions.¹⁴¹ In this structure, the two ends of the thick energy-storage units are designed as cambered surfaces that can effectively buffer the local stress of the interconnected

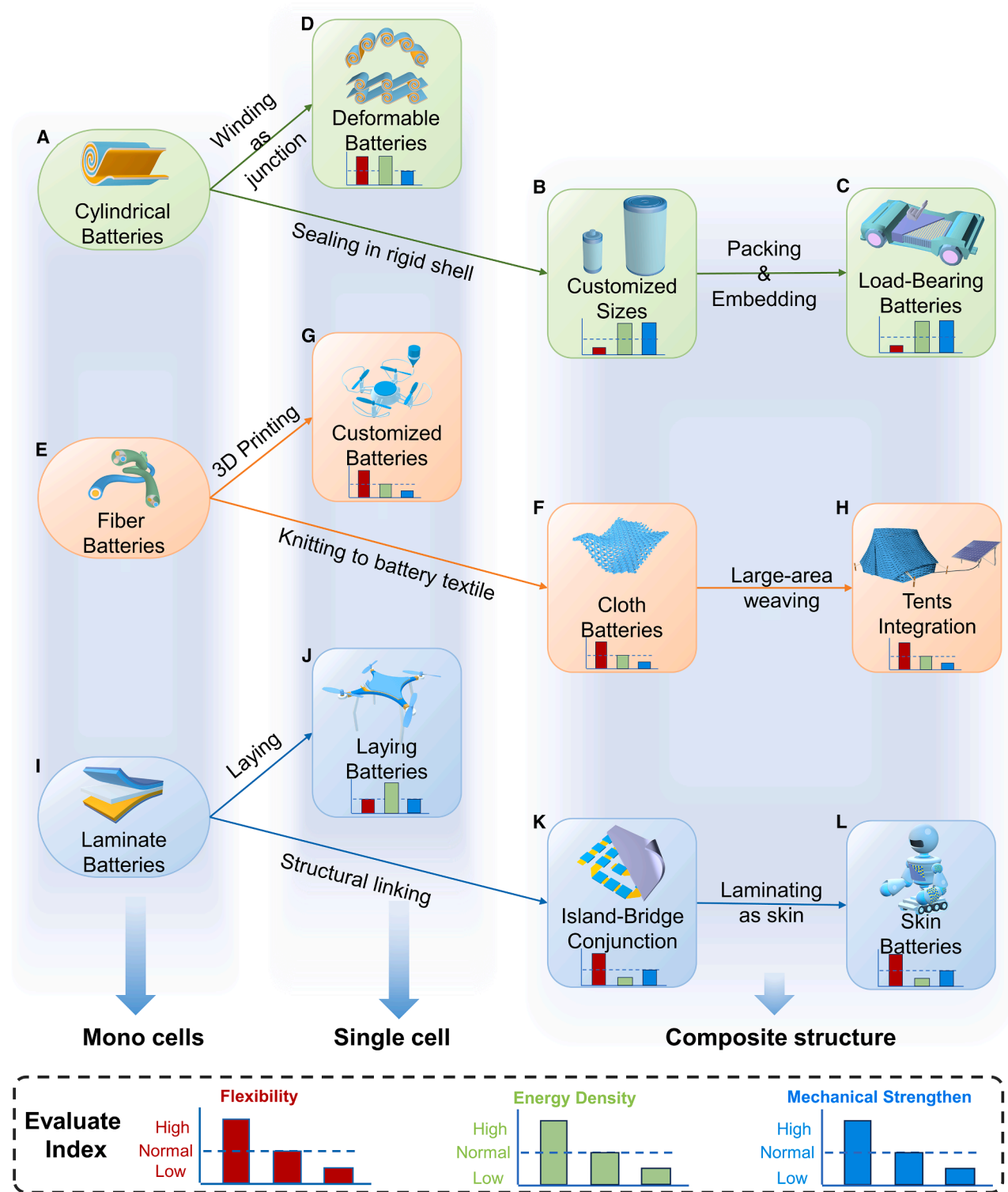


Figure 15. Battery integration strategies based on different types of single cells

(A–D) The evolution of cylindrical single-cell batteries.

(E–H) The evolution of fiber-based batteries.

(I–L) The evolution of laminate-based batteries.

components to realize the mechanical functions of the battery, such as bending, twisting, stretching, and winding. This interesting structural design strategy also provides a choice for the flexible utilization of laminated and deformable winding batteries.

The flexible fabric batteries at the macroscale can be constructed by 3D printing and knitting/weaving single fibers, as shown in Figures 15E–15H.^{142–144} However, long fibers of more than a few centimeters were thought to have higher internal resistances that compromised electrochemical performance. Scalable industrial manufacturing has access to numerous studies to help solve this problem. Peng's group found that the internal resistance of fabricated fiber LIBs has a hyperbolic cotangent function with fiber length, where it first decreases before leveling off as length increases; thus, producing high-performing long-fiber LIBs is possible. Therefore, fiber LIBs were produced by weaving into large-area textiles using an industrial loom, which can produce sufficient energy to power electronic devices. These LIB textiles have been proven to be integrated into everyday clothing for wireless charging of mobile phones and as a real-time health monitor. In addition, a new solution extrusion method can directly and continuously produce entire fiber batteries in a single step at scale with high production rates exceeding 250 m h^{-1} .⁸⁰ A proof of principle for roughly 10 m^2 of woven textile was constructed for a smart self-supply tent application, as shown in Figure 15H; it has an outstanding energy density of 550 mWh m^{-2} . The tent is composed of solar cells (outside) and textile cells (inside), which can obtain energy from the sun during the day and provide energy for electronic devices in the tent at night. As mentioned above, with the advancement of the fabrication means of fiber cells, the scale-up of fiber cell preparation is no longer a bottleneck limiting its development, and the manufacturing cost has been further reduced. Compared with commercial batteries, structural fiber cells have higher environmental adaptability and stimulus response. With advanced applications in energy supply as well as signal and data transmission, fiber-cell-based smart garments will drive the replacement of conventional batteries in next-generation electronics scenarios (Figure 16A).¹⁴⁵

Compared to fiber batteries, laminate batteries have higher energy density, better integrability, and better scalability. As shown in Figures 15I and 15J, flexible laminated batteries have excellent mechanical and electrochemical properties. This kind of laminate battery with conformal function can be used as an electronic shell to lay the structure so as to rationally regulate the space utilization and energy supply. Furthermore, a type of laminate battery modeled after the structure of crocodile skin was constructed to realize bidirectional flexible bending (Figure 16B).¹⁴⁶

Artificial design produces electronics that are appropriate for particular situations; by contrast, biological structures in nature generated from the evolution of organisms are worth imitating in designing functional structures.^{147,148} As shown in Figures 15K and 15L, structurally designed batteries imitating snake scales are assembled using single cells with a certain mechanical strength, which are coupled in a flexible manner to perform both a snake's serpentine movement and robotic joint movement.⁶⁷

Aside from studying the size design of battery packs, the Contemporary Amperex Technology company (CATL) developed an AB battery solution (design solution for managing two different cell moduli, A and B in a single pack) through an advanced battery management system (BMS) for battery systems that can realize the integration, mixing, and sharing of SIBs and LIBs.¹⁴⁹ Both SIBs and LIBs can be simultaneously integrated into one battery system. They can also be mixed and matched according to certain proportions and arrangements, and series and parallel integration can be performed for the balanced control of different battery systems using the precise algorithm of BMS.

Structural batteries are also being explored for a wide range of applications in large transportation vehicles, such as cars, large unmanned aircraft, and airplanes. The main characteristic of structural batteries for large devices is rigidity, which is different from the requirements of small intelligent electronic devices.¹⁵⁰ In a past study, researchers replaced the wings of a macro-air vehicle (MAV) using batteries that can increase the flying time of the MAV by 26%.¹⁵¹ Structural batteries used in aircraft require a radical rethinking of conventional aircraft design to achieve multidisciplinary coupling and synergistic integration with electric propulsion systems for aerodynamics, aeroelasticity, or flight control and then to increase the energy density of storage devices.¹⁵²

Simulations showed that loading A320 aircraft with structural batteries with a power density of more than 200 Wh kg^{-1} resulted in a 15% improvement in energy efficiency during a 1,500-km flight.¹⁵³ Furthermore, a distributed energy system was designed to rationalize the use of space in the passenger cabin of regional aircraft. For the Airbus A220, this system can realize potential mass and volume savings of approximately 260 kg and 510 L while reducing CO_2 and NO_x emissions by approximately 280 and 1.2 tons per year, respectively (Figure 16C).¹⁵⁴

In summary, compared to the commercial batteries in the integrated use of electrified equipment, the designed functional batteries can provide not only sufficient energy supply but also the mechanical strength of the support function.¹⁵³ The increase in range will be achieved by the rational application of the structure, which improves its integrability and space utilization. Moreover, pollution emissions and the cost of aircraft will decrease. Therefore, the design of advanced structural batteries using a combination of different processes and technologies is promising for energy-storage devices in the future.

Modular design and large-format manufacturing

For commercial cylindrical cells or laminate batteries, the integration of battery packs usually has two approaches, namely modular design and large-format manufacturing, corresponding to different advantages and disadvantages. In modular design, these battery packs (usually small cylindrical cells and rectangular prismatic batteries) are designed as interchangeable modules that can be easily disassembled and reassembled. By connecting multiple smaller units, or modules, a larger battery pack with increased capacity or power output can be achieved.¹⁵⁵ The modular design provides flexibility in battery configuration, facilitating easy adaptation to different devices or systems. Moreover, the easy maintenance and repair of modular batteries

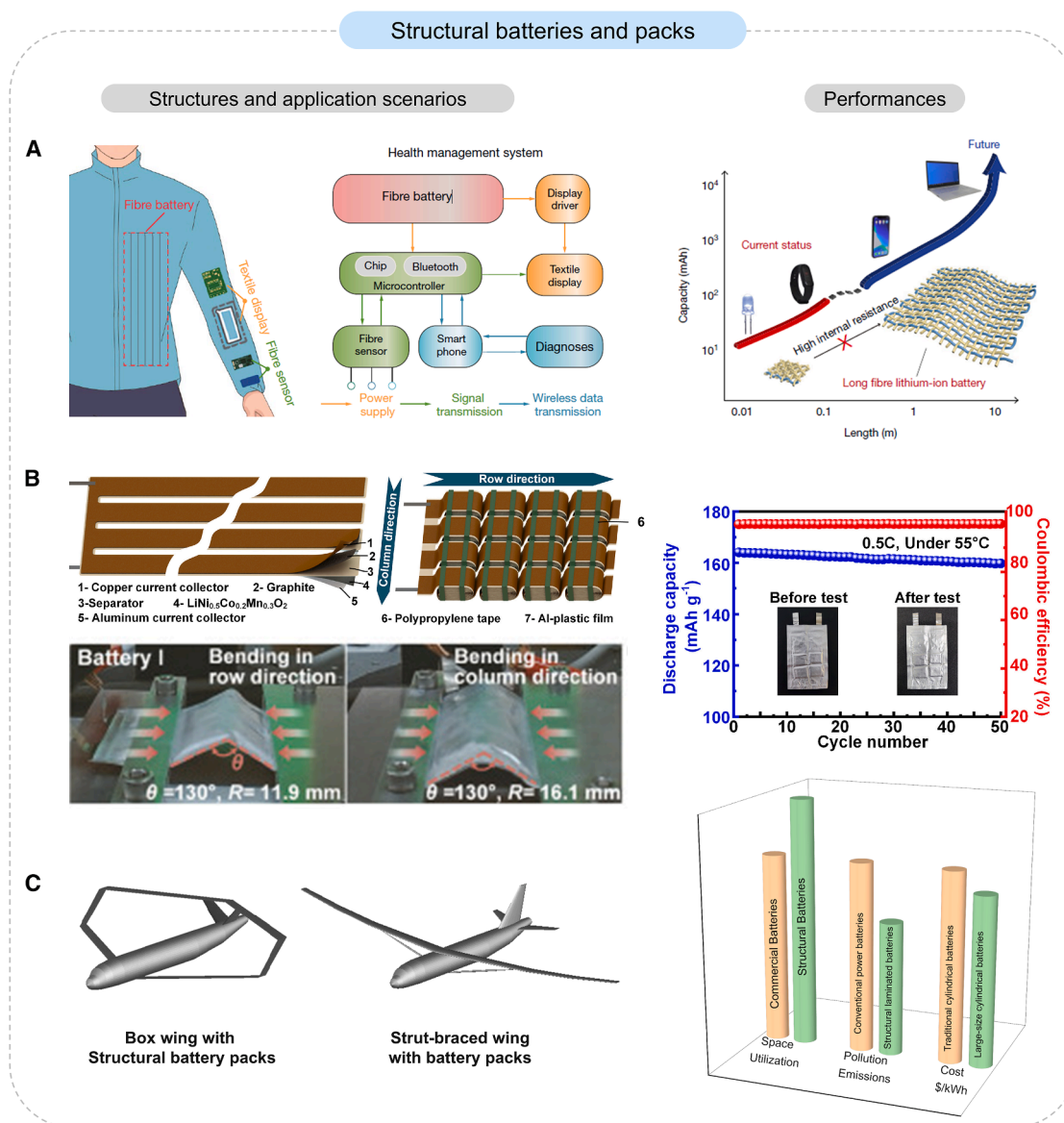


Figure 16. Implications of multifunctional structurally designed battery systems

(A) Structural fiber batteries. Reproduced, with permission, from Zhang et al.¹⁴⁵ Copyright 2024, Springer Nature.

(B) Laminated batteries. Reproduced, with permission, from Liu et al.¹⁴⁶ Copyright 2022, Elsevier.

(C) Structurally designed batteries and the typical advantages of application of structurally designed batteries and packs. Reproduced, with permission, from Karadotcheva et al.¹⁵³ *Energies*, 2021, under CC BY license.

enable individual replacement of faulty or degraded modules, reducing costs and extending the lifespan of the battery system. Nevertheless, the additional connectors and electrical paths between each module add complexity to the design, introduce potential points of failure, and result in a larger overall size, which is a significant disadvantage in applications with limited space.¹⁵⁶

Large-format design can be employed to build a novel battery system with larger physical size and higher power storage compared to the conventional-scale batteries, such as the Tesla 4680 cylindrical battery and BYD blade battery. The integrated design of large-format cells optimizes the internal arrangement

of the battery pack, reduces the design layers, and improves the space utilization of the case, which can achieve higher mass and volumetric energy densities.¹⁵⁷ More significantly, the tab-less design enables direct connection of the positive and negative current collectors to the cover plate and casing, respectively. This configuration substantially increases the current conduction area, shortens the current path length, and reduces internal resistance, thereby markedly enhancing the thermal stability and controllability of the battery. Furthermore, the elongated thin configuration of blade battery cells delivers superior thermal dissipation performance, effectively mitigating

thermal runaway risks and extending service life. Currently, large-format batteries with customized dimensions (mainly large cylindrical cells and large square cells) are widely used in electric vehicles and large-scale energy-storage devices. Furthermore, the integrated design of large-format batteries also tends to attain higher power output, making them adaptable to applications with continuous high current output.¹⁵⁸ However, it is important to note that large-format batteries typically have a fixed size and configuration, which may limit their application in specific spaces of devices. In addition, due to their higher energy loads, the safety of large-format batteries is the main bottleneck of the application.¹⁵⁹

With the demand for higher energy density and more charge loads in power battery packs, large-format designs with better productivity and space utilization are gradually being adopted. In particular, advanced cell-to-pack (CTP) and cell-to-chassis (CTC) technologies are emerging for electric vehicle power battery integration.¹⁶⁰ The mainstream manufacturers with various integration strategies of power packs are summarized in Figure 17. It is found that both CTP and CTC technologies substantially improve the pack space utilization, which is important for enhancing the energy density of automotive power batteries. Notably, the higher-integration cell-to-body (CTB) technology is undergoing rapid development. This approach eliminates the separate casing design characteristic of traditional battery packs, deeply integrating the battery pack into the vehicle body structure. Consequently, the battery pack directly contributes to the overall structural strength of the vehicle.¹⁶¹ Therefore, CTB technology significantly enhances the vehicle body's overall rigidity, with the whole-body torsional stiffness potentially increasing to over 40,000 N·m/².¹⁶² Owing to its structural advantages and performance enhancement potential, CTB technology is anticipated to evolve into a mainstream solution for future automotive power systems.

With the rise of the smart Internet of Things (IoT) era and the increasing popularity of microelectronic devices, the miniaturization of batteries has become an important development trend. However, as the electrode size shrinks ($L_{\text{electrode}}^3$ scaling), the total energy available in microbatteries decreases dramatically; moreover, the reduction in battery size increases the proportion of inactive separator and encapsulation materials, which further reduces the energy density of microbatteries. As a result, existing commercially available microbatteries are only capable of powering an insect-sized crawling robot for about 1 min, which is far from being able to meet the energy requirements of microelectronic devices.¹⁶³ Conventional liquid electrolyte-based LIBs are difficult to fabricate on substrate integration due to size constraints and inherent leakage risks. All-solid-state thin-film lithium/lithium-ion microbatteries are an effective solution, which possess the advantages of all-solid-state battery structure, compatibility with thin-film fabrication process, easy miniaturization, and controllable size, and can be directly integrated into wireless microelectronic devices,¹⁶⁴ which is the focus of future microbattery development.

Compared with commercial cylindrical cells and laminate cells, flexible batteries have greater flexibility in shape, size, and weight and can be customized according to actual needs, providing more design space for wearable devices and flexible

electronic devices. However, at present, flexible batteries are mainly faced with the problems of poor interfacial stability (including the stability of the current collector/active electrode interface and the electrode/electrolyte interface under complex working conditions), complex manufacturing process, and low energy density. Researchers are promoting their development through innovative design of materials and structures. Currently, Tayho's pilot production line for flexible fiber LIBs, in collaboration with Huisheng Peng's team, has been completed and put into production and is undergoing process optimization and commercial promotion.¹⁶⁵ In the next 3–5 years, with the release of pilot production capacity and cost reduction, flexible batteries are expected to achieve large-scale applications in smart wearables and IoT terminals.

MACHINE LEARNING IN MULTIFUNCTIONAL BATTERIES

To accelerate the design and development of structural batteries and reduce the time and cost associated with applying new materials and structures, computational simulation methods are widely employed to guide the optimal design of functional batteries from components to devices. Among these, machine learning (ML), leveraging its powerful capabilities in pattern recognition, high-dimensional optimization, and prediction, is overcoming the efficiency bottleneck of traditional trial-and-error approaches, providing new impetus for material performance prediction and battery structural design. At the micro scale, ML facilitates the efficient screening of target materials by establishing mapping relationships between a material's atomic/molecular structure and its physicochemical properties, thereby predicting its electrochemical properties. Taking solid-state electrolytes as an example, Bradford et al. developed a chemically informed neural network incorporating the Arrhenius equation by combining physical models with data-driven methods to achieve efficient screening of 20,000 solid-state polymer electrolytes.¹⁶⁶ Furthermore, ML molecular dynamics (MLMD) can precisely resolve ion-diffusion behaviors across complex potential energy surfaces, revealing the synergistic mechanism between anionic framework vibrations and lithium-ion migration, thus providing theoretical guidance for designing solid-state electrolytes with high ionic conductivity.¹⁶⁷

For battery packs, thermal management is critical to ensuring normal operation and sustaining high performance. As battery sizes increase, differences in current density and state of charge (SOC) within cells lead to thermal inhomogeneity, thereby affecting the tolerance of the battery pack. Consequently, battery-pack-based thermal management simulations are also critical. By coupling the 3D electrothermal model and artificial neural network model, the influence of different battery-pack configurations (gap parameters between pairs of cells) on cooling efficiency, battery performance, and fire safety has been explored, and the optimized design of a battery thermal management system has been realized.¹⁶⁸ In the future, ML will further integrate physical mechanisms (e.g., electrochemical/thermodynamic equations) to establish closed-loop design across the material-device-system hierarchy, accelerating the development and application of high-safety, high-performance structural batteries.

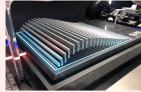
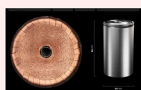
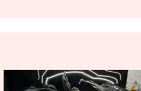
Structural batteries types in vehicles		Cathode Materials	Cell Shape	Stacking Method	Space Utilization	Energy Density	Integration
2020.01	Blade (BYD) 	High Consistency, High Heat Dissipation, High Safety					
		LFP	Long Blade	Stacking	60%	170 Wh/kg	Cell to Pack
2020.03	Beehive (Jianghuai) 	Honeycomb independent structure battery cell design					
		NCM811	Cylinder cell	Winding	60-70%	240 Wh/kg	Cell to Pack
2020.09	4680 (Tesla) 	Tabless cell design, Cylinder size optimization					
		NCM	Cylinder cell	Winding	63%	244 Wh/kg	Cell To Chassis
2021.03	Magazine (Aion) 	Insulated safety compartment, three-dimensional cooling system					
		MCN/LFP	Prismatic Cell	Stacking	50-60%	185/233 Wh/kg	Cell to Pack
2021.06	Dayu (Great Wall) 	Heat source isolation, heat flow distribution					
		NCM811	Prismatic Cell	Stacking	50-60%	190 Wh/kg	Cell to Chassis
2022.05	CTB (BYD) 	Sandwich structure formed by body floor, blade battery, battery tray					
		LFP	Long Blade	Stacking	66%	170 Wh/kg	Cell to Body
2022.09	Amber (VOYAH) 	Three-dementional silicone polymers heat-isolated battery system					
		NCA+SiC	Prismatic Cell	Winding	~50%	/	Cell to Pack
2022.06	Kirin (CATL) 	Buffering, water cooling heat buffering and water cooling					
		LFP/NCM	Prismatic Cell	Stacking	72%	160/255 Wh/kg	Cell to Pack 3.0
2023.04	LongLinJia (SVOLT) 	Bottom cooling outlet to avoid the risk of thermal runaway					
		MFN	Short Blade	Stacking	63.6%	220 Wh/kg	Cell To Chassis
2024.12	CIIC (CATL) 	CIIC can absorb the crash energy of the vehicle					
		LFP/NCM	Short Blade	Stacking	~70%	160/255 Wh/kg	Cell To Chassis

Figure 17. Comparison of power battery pack integration processes for electric vehicles from mainstream manufacturers

CONCLUSIONS AND PERSPECTIVES

Structural design toward functional batteries is drawing substantial attention as advanced functional energy devices that can

enable excellent performance and multifunctional applications through various structural modifications. Here, we systematically summarize recent advances in structural design strategies and important manufacturing processes at the cell, battery, and

pack levels. In this manner, we provide a novel perspective for the development of functional battery systems and the optimization of component materials.

In addition to the basic electrochemical performance and functional requirements, several challenges need to be further studied and optimized to apply structurally designed functional batteries to daily convenient and smart life. These problems are discussed as follows.

- (1) Standardization of dimensions and evaluation. The physical parameters of batteries, such as thickness, diameter, mass, and volume, as well as the inactive components, highly influence performance and application scenarios of battery systems. For example, the diameter of a flexible fiber battery affects its suitability and comfort for textile garments, the size and thickness of a microbattery determine its suitability for smart dust or implantable medical treatment, and the battery management systems integrated in structural batteries decrease the space utilization. In most of the demonstration studies, however, the parameters related to the functional battery performance are not fully given for the researchers to evaluate and compare, which makes the applicability value of the batteries doubtful. At the same time, although some researchers have proposed the necessity of unified testing standards for structurally designed batteries and given recommendations, the functional assessment in some recent works is still diverse or inadequate. Therefore, researchers of structurally designed batteries are expected to unify functional testing standards in the near future and provide comprehensive physical characterizations and detailed key testing parameters.
- (2) Optimization of the manufacturing process. Compared with conventional batteries, structurally designed batteries exhibit more diverse and complex processing methods. The common processing techniques used for structurally designed batteries at different scales need further enhancement for accuracy, continuity, consistency, and production rate, especially for implementation in 3D batteries. Coating technology entails improving the uniformity and interface adhesion of the coatings to ensure even electron/ion transport and prevent misalignment/detachment between battery components. The printing technologies are preferred to develop low-cost and effective ink preparation and reduce the use of polymeric binders or thermoplastic polymer additives because of the obvious damage to electrochemical performance. More compact and streamlined printed batteries are possible with the development of 3D printing technologies with higher resolution and smaller feature sizes for battery manufacture. At the same time, 4D printing technology allows structural or functional changes over time, providing a potential solution for developing responsive and editable batteries. In addition, large-scale production poses a great challenge to the commercialization and promotion of structurally designed batteries. At this stage, except for the more researched structurally designed batteries with special physical functions, most

structural designs are concept demonstrations, and systematic large-scale manufacturing research is still lacking.

- (3) Reduction of electrochemistry performance sacrifice. As already mentioned, low active material mass loading, high percentage of inactive functional materials, low ionic conductivity of gel/solid electrolyte, and large ion-transport distance between electrodes often were apparent in battery functionalization, leading the structurally designed batteries to easily exhibit electrochemical drawbacks of low energy/power density and short cycle life. Balancing the functionality and electrochemical performance of structural designs is an important topic. The solutions are considered from material optimization, structural design, and fabrication processes, such as developing low-cost and efficient functional materials such as CNT and graphene paper; using anticorrosive collectors as battery encapsulation materials simultaneously; building structurally designed batteries to reduce binder introduction; designing self-assembled structures to avoid inactive porous templates; and directly printing active material slurry without additives.
- (4) Improvement in the safety of the device. Safety is the prerequisite for the structurally designed battery to be used in real life. The special structure of structurally designed batteries can lead to idiosyncrasies in internal electron/ion transport, interface contact, heat dissipation, and sealing, which in turn affect the safe operation of the battery. For example, long-fiber batteries usually have a high internal resistance, which poses the safety hazard of serious heat generation under high power operation. Incomplete interface contact in the deformed state can also lead to uneven charge distribution, damaging the electrodes and causing thermal runaway. In addition, structural batteries may face varying degrees of vibration, deformation, extrusion, and other harsh conditions during operation and are vulnerable to the leakage of toxic substances, of which the flammable electrolyte battery system carries a higher safety risk. Regarding these issues, researchers can consider using aqueous electrolytes, non-flammable electrolytes, semi-solid/solid electrolytes, flame-retardant collectors/separators, and self-healing electrode materials, optimizing the battery-sealing process to prevent leakage, integrating battery management chips to monitor battery operation status, and increasing battery system safety testing.

Finally, starting from existing research and development needs, to fully realize the potential of structurally designed functional batteries, we outline the evolutionary road map of four development generations based on increasing levels of versatility, integration, and intelligence, as illustrated in [Figure 18](#). The critical materials, key techniques, and indicators for four generations are also listed.

The first generation comprises single-functional structurally designed batteries. The initial generation focuses on implementing a single structural or functional enhancement into conventional battery systems. Leveraging mature fabrication techniques (e.g.,

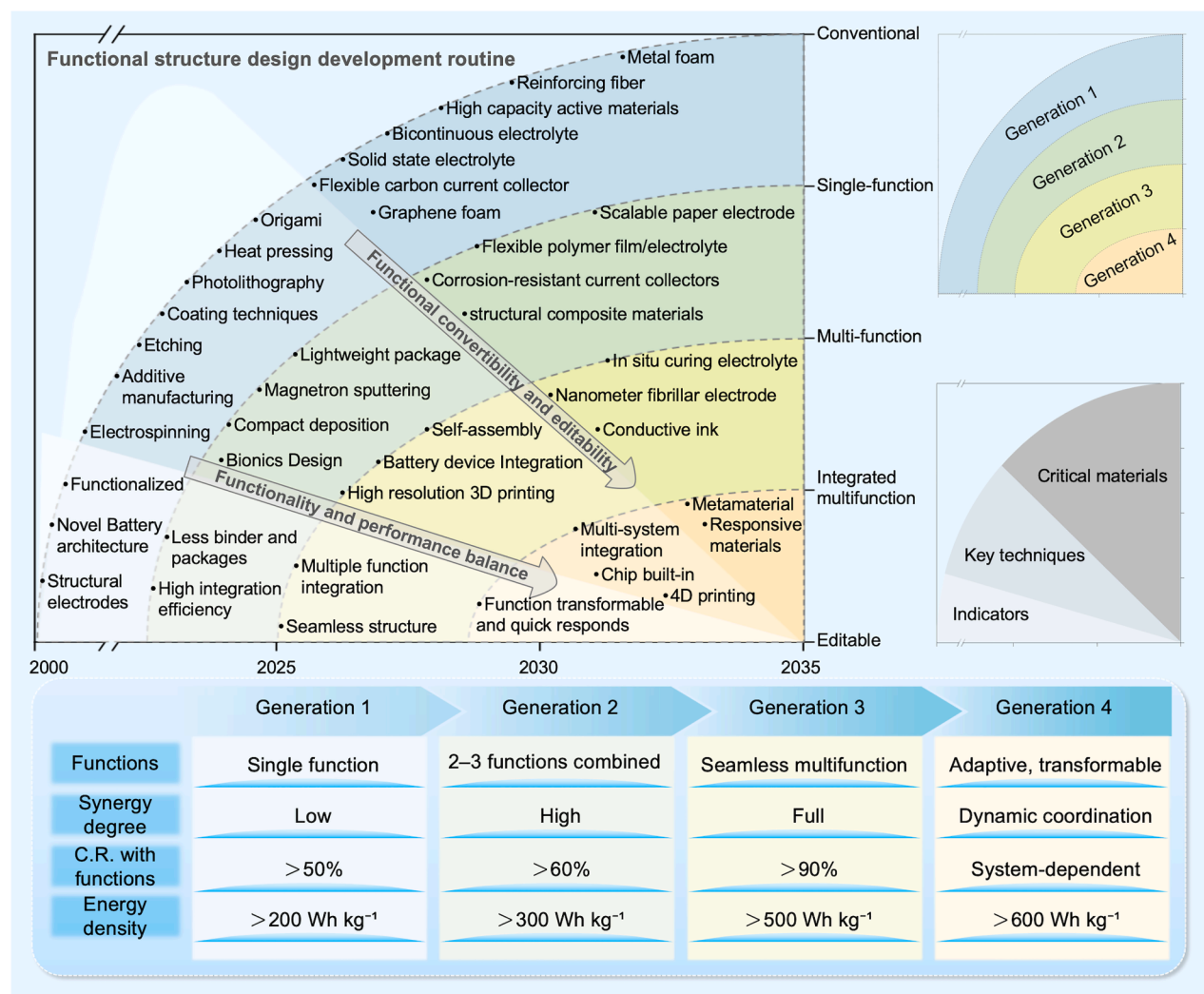


Figure 18. Development road map for functional batteries with structural design

Key techniques, critical materials, and indicators for each generation of functional battery.

coating, etching, and electrospinning) and existing functional materials (e.g., graphene foam, metal foam, and carbon paper), this generation mainly targets structural electrodes, novel architectures, or basic integration schemes to endow batteries with a specific property (e.g., flexibility or lightweight packaging). Most of the currently reported structurally designed functional batteries fall into this generation. The synergy degree between function and electrochemical performance remains low, with the expectation that, after functionalization, the battery retains over 50% of the capacity compared to its non-functionalized counterpart and reaches an energy density above 200 Wh kg⁻¹.

The second generation comprises multifunctional structurally designed batteries. Multiple functions are combined into a single battery, enabling broader application scenarios and addressing more complex demands. Techniques such as compact deposition, bionic design, and lightweight package begin to support this level of integration. There have been some studies on key function integration, such as flexible all-in-one batteries and

transparent flexible batteries. Although the synergy degree is improved, careful assessment of functional compatibility and structural rationality remains essential. The targeted capacity retention (CR) after functionalization is increased to 60%, along with an energy density of 300 Wh kg⁻¹, setting a new benchmark for multifunctional battery designs.

The third generation comprises all-in-one multifunctional structurally designed batteries. Building upon the second generation, the third generation pursues seamless multifunctional integration by employing advanced materials (e.g., *in situ* curing electrolytes and conductive ink) and system-level integration strategies (e.g., self-assembly and high-resolution 3D printing). Functions are not only coexisting but mutually enhancing, enabling compact structures with reduced inactive components and near-complete synergy between functionality and electrochemical performance. The target CR with functions and energy density for this generation are 90% and 500 Wh kg⁻¹, respectively, meeting the demands of high-performance multifunctional devices.

The fourth generation comprises dynamically editable functional batteries. This generation envisions adaptive, reconfigurable, and intelligent batteries. With the use of responsive materials, metamaterials, and technologies such as 4D printing and multisystem integration, batteries can actively respond to environmental changes, transforming their function and structure on demand. Recent studies have reported the use of rapid thermal-switching responsive materials as cell-to-cell interlayers to enhance the operational safety of battery packs.¹⁶⁹ In addition, 4D printing technology has been employed to chemically integrate liquid crystal elastomers with flexible zinc-ion microbatteries, effectively addressing the interfacial delamination issues found in traditional stacked configurations.¹⁷⁰ The dynamic coordination design across multiple system levels, enabled by advanced materials and techniques at different scales, will be key to the development of this generation of functional batteries. To ensure that adaptivity does not compromise the fundamental electrochemical performance, the system is expected to achieve an energy density typically above 600 Wh kg⁻¹.

In addition to the development of materials and technology, it is also important to recognize that different battery chemistries inherently possess distinct advantages and limitations when applied to multifunctional systems. For large-scale energy storage, SIBs are attracting attention due to abundant resources and low cost of metallic Na, although improvements in cycle life and energy density are still needed. Vanadium redox flow batteries, with their excellent safety and scalability, are promising for modular and reconfigurable energy systems. Aqueous batteries, particularly aqueous ZIBs, are highly suited to wearable and implantable medical devices due to their intrinsic safety, biocompatibility, and ability to integrate with flexible gel electrolytes and high-capacity cathodes. Solid-state lithium batteries offer enhanced safety, mechanical strength, and energy density due to the non-flammable solid electrolytes and stable interfaces, making them attractive for structural batteries in advanced portable electronics and electric vehicles requiring high reliability. Organic batteries, characterized by their tunable molecular structures and environmental friendliness, provide unique opportunities for flexible and biodegradable energy-storage devices. For extreme environmental applications such as spacecraft, geological prospectors, and polar research vessels, Mg metal batteries with excellent wide-temperature-range performance, stability, and volume energy density are a promising solution. The structural design strategies are expected to be coordinated with the underlying electrochemical system to fully realize its functional potential.

Functional batteries exhibit far superior electrochemical performance and potential for application in multiple scenarios. Studies on structural design strategies will further explore innovative inner battery structures, multifunctional fusion, and efficient battery packs as well as drive the development of precision and efficiency in manufacturing processes. The demand for specialized functional batteries in smart cities will accelerate their development, which will eventually broaden the field of study and open new markets for application.

ACKNOWLEDGMENTS

This work was supported by the Joint Funds of the National Natural Science Foundation of China (U2130204), the National Key R&D Program of China (2022YFB2502102), the Beijing Outstanding Young Scientists Program (BJJWZYJH01201910007023), and the Young Elite Scientists Sponsorship Program by CAST (YESS20200364).

Q7

AUTHOR CONTRIBUTIONS

Conceptualization, R.C., Y.H., Y.M., and F.W.; writing – original draft, Y.M., X. H., and H.W.; writing – review & editing, Y.M., L.L., and Y.H.; supervision, R.C., Y.H., and Y.M.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT in order to check grammar and polish expressions. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

REFERENCES

- Scrosati, B., Hassoun, J., and Sun, Y.-K. (2011). Lithium-ion batteries. A look into the future. *Energy Environ. Sci.* 4, 3287–3295.
- Li, M., Lu, J., Chen, Z., and Amine, K. (2018). 30 years of lithium-ion batteries. *Adv. Mater.* 30, 1800561.
- Xie, J., and Lu, Y.-C. (2020). A retrospective on lithium-ion batteries. *Nat. Commun.* 11, 2499.
- Rodrigues, M.-T.F., Babu, G., Gullapalli, H., Kalaga, K., Sayed, F.N., Kato, K., Joyner, J., and Ajayan, P.M. (2017). A materials perspective on Li-ion batteries at extreme temperatures. *Nat. Energy* 2, 17108–17114.
- Mallarapu, A., Kim, J., Carney, K., DuBois, P., and Santhanagopalan, S. (2020). Modeling extreme deformations in lithium ion batteries. *ETransportation* 4, 100065.
- Snyder, J., Baechle, D., Wetzel, E., and Xu, K. (2008). Multifunctional structural composite batteries for US army applications. *Proc. Army Sci. Conf.* (26th), 1–8.
- Asp, L.E., and Greenhalgh, E.S. (2014). Structural power composites. *Compos. Sci. Technol.* 101, 41–61.
- Asp, L.E., Johansson, M., Lindbergh, G., Xu, J., and Zenkert, D. (2019). Structural battery composites: a review. *Funct. Compos. Struct.* 1, 042001.
- Wang, D., Hwang, J., Chen, C.y., Kubota, K., Matsumoto, K., and Hagiwara, R. (2021). A β'' -Alumina/Inorganic Ionic Liquid Dual Electrolyte for Intermediate-Temperature Sodium-Sulfur Batteries. *Adv. Funct. Mater.* 31, 2105524.
- Liu, Q., Zhang, G., Chen, N., Feng, X., Wang, C., Wang, J., Jin, X., and Qu, L. (2020). The First Flexible Dual-Ion Microbattery Demonstrates Superior Capacity and Ultrahigh Energy Density: Small and Powerful. *Adv. Funct. Mater.* 30, 2002086.
- Qian, G., Liao, X., Zhu, Y., Pan, F., Chen, X., and Yang, Y. (2019). Designing flexible lithium-ion batteries by structural engineering. *ACS Energy Lett.* 4, 690–701.
- Wang, M., Emre, A., Tung, S., Gerber, A., Wang, D., Huang, Y., Cecen, V., and Kotov, N.A. (2019). Biomimetic solid-state Zn²⁺ electrolyte for corrugated structural batteries. *ACS Nano* 13, 1107–1115.

13. Larché, F.C., and Cahn, J.W. (1985). Overview no. 41 The interactions of composition and stress in crystalline solids. *Acta Metall.* 33, 331–357.
14. Li, Z., Wu, Z., Wu, S., Tang, W., Qiu, J., Liu, T., Lin, Z., and Lu, J. (2024). Designing Advanced Polymeric Binders for High-Performance Rechargeable Sodium Batteries. *Adv. Funct. Mater.* 34, 2307261.
15. Fan, Z., Chen, X., Shi, J., Nie, H., Zhang, X., Zhou, X., Xie, X., and Xue, Z. (2025). Functionalized Separators Boosting Electrochemical Performances for Lithium Batteries. *Nano-Micro Lett.* 17, 128.
16. Xie, H., Han, B., Song, H., Li, X., Kang, Y., and Zhang, Q. (2021). In-situ measurements of electrochemical stress/strain fields and stress analysis during an electrochemical process. *J. Mech. Phys. Solids* 156, 104602.
17. Choi, J., Polyzos, G., Humphrey, H.E., Toomey, M., Kanbargi, N., Naskar, A., Belharouak, I., and Sharma, J. (2025). Long carbon fibers boost performance of dry processed Li-ion battery electrodes. *J. Power Sources* 640, 236603.
18. Niu, X., Lu, Y., Chen, P., Cao, C., Liao, X., He, L., and Ni, Y. (2025). Gradient-Matched Microstructural Engineering for Fast-Charging, Damage-Tolerant Thick Electrodes of Lithium-Ion Batteries. *Adv. Energy Mater.* 15, 2502245.
19. Kuo, J.J., Kang, S.D., and Chueh, W.C. (2022). Contact Resistance of Carbon-Li_x(Ni,Mn,Co)O₂ Interfaces. *Adv. Energy Mater.* 12, 2201114.
20. Xia, J., Niu, Y., Yin, S., Cong, L., Yu, J., Zhou, H., Guan, X., Xiong, G., Lu, Q., Zhang, S., et al. (2023). Lamellar multi-arch microstructure of Co₃O₄/N-CNTs-CNF composites as anodes for high-performance flexible lithium-ion batteries. *J. Energy Storage* 74, 109481.
21. Chaudhary, R., Xu, J., Xia, Z., and Asp, L.E. (2024). Unveiling the Multifunctional Carbon Fiber Structural Battery. *Adv. Mater.* 36, 2409725.
22. Mei, W., Duan, Q., Qin, P., Xu, J., Wang, Q., and Sun, J. (2019). A three-dimensional electrochemical-mechanical model at the particle level for lithium-ion battery. *J. Electrochem. Soc.* 166, A3319–A3331.
23. Wang, Q., Xiang, L., Lu, H., Liu, S., Hao, C., Liu, J., Han, J., Yang, J., NuLi, Y., and Wang, J. (2025). Energy-dissipating binder for self-healing sulfurized polyacrylonitrile electrode for rechargeable lithium batteries. *Chem. Eng. J.* 517, 164296.
24. Song, Y., Xu, S., Sato, S., Lee, I., Xu, X., Omori, T., Nagasako, M., Kawasaki, T., Kiyanagi, R., Harjo, S., et al. (2025). A lightweight shape-memory alloy with superior temperature-fluctuation resistance. *Nature* 638, 965–971.
25. Jin, T., Singer, G., Liang, K., and Yang, Y. (2023). Structural batteries: Advances, challenges and perspectives. *Mater. Today* 62, 151–167.
26. Danzi, F., Salgado, R.M., Oliveira, J.E., Arteiro, A., Camanho, P.P., and Braga, M.H. (2021). Structural batteries: A review. *Molecules* 26, 2203.
27. Wang, G., Yu, M., and Feng, X. (2021). Carbon materials for ion-intercalation involved rechargeable battery technologies. *Chem. Soc. Rev.* 50, 2388–2443.
28. Wang, H., Zhou, A., Hu, X., Hu, Z., Zhang, F., Huang, Y., Li, L., Wu, F., and Chen, R. (2023). Bifunctional Dynamic Adaptive Interphase Reconfiguration for Zinc Deposition Modulation and Side Reaction Suppression in Aqueous Zinc Ion Batteries. *ACS Nano* 17, 1557–1589.
29. Jin, T., Ma, Y., Xiong, Z., Fan, X., Luo, Y., Hui, Z., Chen, X., and Yang, Y. (2021). Bioinspired, tree-root-like interfacial designs for structural batteries with enhanced mechanical properties. *Adv. Energy Mater.* 11, 2100997.
30. Asp, L.E., Bouton, K., Carlstedt, D., Duan, S., Harnden, R., Johansson, W., Johansen, M., Johansson, M.K.G., Lindbergh, G., Liu, F., et al. (2021). A structural battery and its multifunctional performance. *Adv. Energy Sustain. Res.* 2, 2170008.
31. Bouton, K., Schneider, L., Zenkert, D., and Lindbergh, G. (2024). A structural battery with carbon fibre electrodes balancing multifunctional performance. *Compos. Sci. Technol.* 256, 110728.
32. Carlstedt, D., Rittweger, F., Runesson, K., Navarro-Suárez, A.M., Xu, J., Duan, S., Larsson, F., Riemschneider, K.-R., and Asp, L.E. (2022). Experimental and computational characterization of carbon fibre based structural battery electrode laminae. *Compos. Sci. Technol.* 220, 109283.
33. Ma, Y., Li, L., Qian, J., Qu, W., Luo, R., Wu, F., and Chen, R. (2021). Materials and structure engineering by magnetron sputtering for advanced lithium batteries. *Energy Storage Mater.* 39, 203–224.
34. Pinilla, S., Ryan, S., McKeon, L., Lian, M., Vaesen, S., Roy, A., Schmitt, W., Coleman, J.N., and Nicolosi, V. (2023). Additive Manufacturing of Li-Ion Batteries: A Comparative Study between Electrode Fabrication Processes. *Adv. Energy Mater.* 13, 2203747.
35. Zhang, W., Tang, H., Yan, Y., Ma, J., Ferro, L.M.M., Mercedes, L., Kar-naushenko, D.D., Karnaushenko, D., Schmidt, O.G., and Zhu, M. (2024). Unlocking Micro-Origami Energy Storage. *ACS Appl. Energy Mater.* 7, 11256–11268.
36. Huang, Y., Ip, W.S., Lau, Y.Y., Sun, J., Zeng, J., Yeung, N.S.S., Ng, W.S., Li, H., Pei, Z., Xue, Q., et al. (2017). Weavable, conductive yarn-based NiCo/Zn textile battery with high energy density and rate capability. *ACS Nano* 11, 8953–8961.
37. Lyu, Z., Koh, J.J., Lim, G.J.H., Zhang, D., Xiong, T., Zhang, L., Liu, S., Duan, J., Ding, J., Wang, J., et al. (2022). Direct ink writing of programmable functional silicone-based composites for 4D printing applications. *Interdiscip. Mater.* 1, 507–516.
38. Saari Luoma, H., Piironen, A., Unt, A., Hakanen, J., Rautava, T., and Salminen, A. (2020). Overview of optical digital measuring challenges and technologies in laser welded components in EV battery module design and manufacturing. *Batteries* 6, 47.
39. Peng, H.-J., Huang, J.-Q., and Zhang, Q. (2017). A review of flexible lithium-sulfur and analogous alkali metal-chalcogen rechargeable batteries. *Chem. Soc. Rev.* 46, 5237–5288.
40. Shadike, Z., Lee, H., Borodin, O., Cao, X., Fan, X., Wang, X., Lin, R., Bak, S.M., Ghose, S., Xu, K., et al. (2021). Identification of LiH and nanocrystalline LiF in the solid-electrolyte interphase of lithium metal anodes. *Nat. Nanotechnol.* 16, 549–554.
41. Dong, H., Li, J., Guo, J., Lai, F., Zhao, F., Jiao, Y., Brett, D.J.L., Liu, T., He, G., and Parkin, I.P. (2021). Insights on Flexible Zinc-Ion Batteries from Lab Research to Commercialization. *Adv. Mater.* 33, 2007548.
42. Rus, D., and Tolley, M.T. (2015). Design, fabrication and control of soft robots. *Nature* 521, 467–475.
43. Luo, C., Wen, S., Hu, H., Tang, Q., Chi, S.-S., Wang, J., Huang, W., Yang, Y., Wang, C., Deng, Y., and Chang, J. (2023). Bamboo mat-inspired interlocking compact textile electrodes for high-energy-density flexible lithium-ion full batteries. *Energy Storage Mater.* 55, 388–396.
44. Zhang, P., Wang, K., Zuo, Y., Wei, M., Pei, P., Liu, J., Wang, H., Chen, Z., and Shang, N. (2022). A flexible zinc-air battery using fiber absorbed electrolyte. *J. Power Sources* 531, 231342.
45. Li, Y., Guan, Q., Cheng, J., and Wang, B. (2022). Amorphous H₀.82MoO₃. 26 cathodes based long cyclife fiber-shaped Zn-ion battery for wearable sensors. *Energy Storage Mater.* 49, 227–235.
46. Wang, Y., Chen, C., Xie, H., Gao, T., Yao, Y., Pastel, G., Han, X., Li, Y., Zhao, J., Fu, K.K., and Hu, L. (2017). 3D-printed all-fiber Li-ion battery toward wearable energy storage. *Adv. Funct. Mater.* 27, 1703140.
47. Rao, Y., Chen, S., Yue, Q., and Kang, Y. (2021). Optimizing the spin states of mesoporous Co₃O₄ nanorods through vanadium doping for long-lasting and flexible rechargeable Zn-air batteries. *ACS Catal.* 11, 8097–8103.
48. Wu, Z., Liu, K., Lv, C., Zhong, S., Wang, Q., Liu, T., Liu, X., Yin, Y., Hu, Y., Wei, D., and Liu, Z. (2018). Ultrahigh-energy density lithium-ion cable battery based on the carbon-nanotube woven macrofilms. *Small* 14, 1800414.
49. Wang, X., Pan, Z., Yang, J., Lyu, Z., Zhong, Y., Zhou, G., Qiu, Y., Zhang, Y., Wang, J., and Li, W. (2019). Stretchable fiber-shaped lithium metal anode. *Energy Storage Mater.* 22, 179–184.
50. Zhao, Z., Wang, R., Peng, C., Chen, W., Wu, T., Hu, B., Weng, W., Yao, Y., Zeng, J., Chen, Z., et al. (2021). Horizontally arranged zinc platelet electrode deposits modulated by fluorinated covalent organic framework

- film for high-rate and durable aqueous zinc ion batteries. *Nat. Commun.* **12**, 6606.
51. Liu, G., Kim, J.Y., Wang, M., Woo, J.Y., Wang, L., Zou, D., and Lee, J.K. (2018). Soft, highly elastic, and discharge-current-controllable eutectic gallium–indium liquid metal–air battery operated at room temperature. *Adv. Energy Mater.* **8**, 1703652.
52. Senthil, C., Kim, S.-S., and Jung, H.Y. (2022). Flame retardant high-power Li-S flexible batteries enabled by bio-macromolecular binder integrating conformational fractions. *Nat. Commun.* **13**, 145.
53. Xu, M., Dou, H., Zhang, Z., Zheng, Y., Ren, B., Ma, Q., Wen, G., Luo, D., Yu, A., Zhang, L., et al. (2022). Hierarchically Nanostructured Solid-State Electrolyte for Flexible Rechargeable Zinc-Air Batteries. *Angew. Chem. Int. Ed.* **61**, e202117703.
54. Yin, L., Cao, M., Kim, K.N., Lin, M., Moon, J.-M., Sempionatto, J.R., Yu, J., Liu, R., Wicker, C., Trifonov, A., et al. (2022). A stretchable epidermal sweat sensing platform with an integrated printed battery and electrochromic display. *Nat. Electron.* **5**, 694–705.
55. Chen, X., Huang, H., Pan, L., Liu, T., and Niederberger, M. (2019). Fully integrated design of a stretchable solid-state lithium-ion full battery. *Adv. Mater.* **31**, 1904648.
56. Zhu, H.-W., Ge, J., Peng, Y.-C., Zhao, H.-Y., Shi, L.-A., and Yu, S.-H. (2018). Dip-coating processed sponge-based electrodes for stretchable Zn-MnO₂ batteries. *Nano Res.* **11**, 1554–1562.
57. Li, H., Ding, Y., Ha, H., Shi, Y., Peng, L., Zhang, X., Ellison, C.J., and Yu, G. (2017). An all-stretchable-component sodium-ion full battery. *Adv. Mater.* **29**, 1700898.
58. Karami-Mosammam, M., Danninger, D., Schiller, D., and Kaltenbrunner, M. (2022). Stretchable and biodegradable batteries with high energy and power density. *Adv. Mater.* **34**, 2204457.
59. Qu, S., Liu, B., Wu, J., Zhao, Z., Liu, J., Ding, J., Han, X., Deng, Y., Zhong, C., and Hu, W. (2020). Kirigami-inspired flexible and stretchable zinc-air battery based on metal-coated sponge electrodes. *ACS Appl. Mater. Interfaces* **12**, 54833–54841.
60. Choi, S., Lee, D., Kim, G., Lee, Y.Y., Kim, B., Moon, J., and Shim, W. (2017). Shape-reconfigurable aluminum–air batteries. *Adv. Funct. Mater.* **27**, 1702244.
61. Zhao, J., Hu, H., Fang, W., Bai, Z., Zhang, W., and Wu, M. (2021). Liquid-metal-bridge~ island design: seamless integration of intrinsically stretchable liquid metal circuits and mechanically deformable structures for ultra-stretchable all-solid-state rechargeable Zn–air battery arrays. *J. Mater. Chem. A* **9**, 5097–5110.
62. Xu, S., Zhang, Y., Cho, J., Lee, J., Huang, X., Jia, L., Fan, J.A., Su, Y., Su, J., Zhang, H., et al. (2013). Stretchable batteries with self-similar serpentine interconnects and integrated wireless recharging systems. *Nat. Commun.* **4**, 1543.
63. Qu, S., Liu, J., Han, X., Deng, Y., Zhong, C., and Hu, W. (2022). Dynamic stretching–electroplating metal-coated textile for a flexible and stretchable zinc–air battery. *Carbon Energy* **4**, 867–877.
64. Kwon, O.H., Oh, J.H., Gu, B., Jo, M.S., Oh, S.H., Kang, Y.C., Kim, J.K., Jeong, S.M., and Cho, J.S. (2020). Porous SnO₂/C nanofiber anodes and LiFePO₄/C nanofiber cathodes with a wrinkle structure for stretchable lithium polymer batteries with high electrochemical performance. *Adv. Sci.* **7**, 2001358.
65. Liu, J., Hu, M., Wang, J., Nie, N., Wang, Y., Wang, Y., Zhang, J., and Huang, Y. (2019). An intrinsically 400% stretchable and 50% compressible NiCo/Zn battery. *Nano Energy* **58**, 338–346.
66. Gonzalez-Guerrero, M., and Gomez, F.A. (2019). An all-printed 3D-Zn/Fe₃O₄ paper battery. *Sens. Actuators B Chem.* **289**, 226–233.
67. Song, Z., Ma, T., Tang, R., Cheng, Q., Wang, X., Krishnaraju, D., Panat, R., Chan, C.K., Yu, H., and Jiang, H. (2014). Origami lithium-ion batteries. *Nat. Commun.* **5**, 3140.
68. Praveen, S., Sim, G.S., Ho, C.W., and Lee, C.W. (2021). 3D-printed twisted yarn-type Li-ion battery towards smart fabrics. *Energy Storage Mater.* **41**, 748–757.
69. Wang, C., Zhang, Y., Liao, M., Li, P., Ma, L., Yao, H., Qu, J., Zhang, K., Li, C., Song, T., et al. (2025). A Novel Coating-Extrusion Method Enabled, High Energy, Power Density, and Scalable Production in Monolithically Integrated Energy Storage Fibers. *Adv. Mater.* **37**, 2417662.
70. Khudiyev, T., Grena, B., Loke, G., Hou, C., Jang, H., Lee, J., Noel, G.H., Alain, J., Joannopoulos, J., Xu, K., et al. (2022). Thermally drawn rechargeable battery fiber enables pervasive power. *Mater. Today* **52**, 80–89.
71. Wang, Z., Ma, Q., Wu, Z., Zhou, J., Zhang, Q., Zhang, Q., Lv, P., Yu, K., and Wei, W. (2025). Thermal and ionic conductivity improvement of LAGP glass ceramic and fiber thermal drawing. *Ceram. Int.* **51**, 8615–8626.
72. Loke, G., Yan, W., Khudiyev, T., Noel, G., and Fink, Y. (2020). Recent Progress and Perspectives of Thermally Drawn Multimaterial Fiber Electronics. *Adv. Mater.* **32**, 1904911.
73. Li, Y., Wang, Y., Liu, Y., Lin, Z., Ye, Z., Wang, J., Qiu, J., Zhang, H., and Cao, G. (2023). In-situ interface reinforcement for 3D printed fiber electrodes. *Energy Storage Mater.* **57**, 497–507.
74. Ma, L., Chen, S., Wang, D., Yang, Q., Mo, F., Liang, G., Li, N., Zhang, H., Zapfen, J.A., and Zhi, C. (2019). Super-Stretchable Zinc–Air Batteries Based on an Alkaline-Tolerant Dual-Network Hydrogel Electrolyte. *Adv. Energy Mater.* **9**, 1803046.
75. Li, L., Chen, Y., Wang, S., Pei, D., Li, M., Li, T., and Li, C. (2024). Coaxially spinning stretchable zin-ion battery fiber with waterproof and scissorability. *Nano Energy* **126**, 109662.
76. Li, C., Ye, Q., Wang, J., Huang, X., Song, T., Zhang, K., Li, P., Zhang, Y., Gong, X., Jiang, Y., et al. (2025). Ultrathin and capacity-tunable lithium metal wires for lithium-based fiber batteries. *Natl. Sci. Rev.* **12**, nwae480.
77. Xu, Q., Chen, J., Loh, J.R., Zhong, H., Zhang, K., Xue, J., and Lee, W.S.V. (2024). Fiber-Shaped Batteries Towards High Performance and Perspectives of Corresponding Integrated Battery Textiles. *Adv. Energy Mater.* **14**, 2302536.
78. Li, Y., Wang, Y., Liu, Y., Yan, F., Zhu, Z., Chen, X., Qiu, J., Zhang, H., and Cao, G. (2024). Polymer Engineering Enables High Linear Capacity Fiber Electrodes by Microenvironment Regulation. *Adv. Sci.* **11**, 2309461.
79. Mo, F., Liang, G., Huang, Z., Li, H., Wang, D., and Zhi, C. (2020). An Overview of Fiber-Shaped Batteries with a Focus on Multifunctionality, Scalability, and Technical Difficulties. *Adv. Mater.* **32**, 1902151.
80. Liao, M., Wang, C., Hong, Y., Zhang, Y., Cheng, X., Sun, H., Huang, X., Ye, L., Wu, J., Shi, X., et al. (2022). Industrial scale production of fibre batteries by a solution-extrusion method. *Nat. Nanotechnol.* **17**, 372–377.
81. Chen, S., Zhang, J., Wang, Z., Nie, L., Hu, X., Yu, Y., and Liu, W. (2021). Electrocatalytic NiCo₂O₄ nanofiber arrays on carbon cloth for flexible and high-loading lithium–sulfur batteries. *Nano Lett.* **21**, 5285–5292.
82. Chen, S., Tao, R., Tu, J., Guo, P., Yang, G., Wang, W., Liang, J., and Lu, S.Y. (2021). High Performance Flexible Lithium-Ion Battery Electrodes: Ion Exchange Assisted Fabrication of Carbon Coated Nickel Oxide Nanosheet Arrays on Carbon Cloth. *Adv. Funct. Mater.* **31**, 2101199.
83. Wang, C.-Y., Zheng, Z.-J., Feng, Y.-Q., Ye, H., Cao, F.-F., and Guo, Z.-P. (2020). Topological design of ultrastrong MXene paper hosted Li enables ultrathin and fully flexible lithium metal batteries. *Nano Energy* **74**, 104817.
84. Song, W.J., Kong, M., Cho, S., Lee, S., Kwon, J., Son, H.B., Song, J.H., Lee, D.G., Song, G., Lee, S.Y., et al. (2020). Stand-alone intrinsically stretchable electronic device platform powered by stretchable rechargeable battery. *Adv. Funct. Mater.* **30**, 2003608.
85. Tian, H., Yang, J.L., Deng, Y., Tang, W., Liu, R., Xu, C., Han, P., and Fan, H.J. (2022). Steel Anti-Corrosion Strategy Enables Long-Cycle Zn Anode. *Adv. Energy Mater.* **13**, 202202603.
86. Shen, Q., Jiang, M., Wang, R., Song, K., Vong, M.H., Jung, W., Krisnadi, F., Kan, R., Zheng, F., Fu, B., et al. (2023). Liquid metal-based soft,

- hermetic, and wireless-communicable seals for stretchable systems. *Science* **379**, 488–493.
87. Song, Z., Wang, X., Lv, C., An, Y., Liang, M., Ma, T., He, D., Zheng, Y.-J., Huang, S.-Q., Yu, H., and Jiang, H. (2015). Kirigami-based stretchable lithium-ion batteries. *Sci. Rep.* **5**, 10988.
88. Li, N., Chen, H., Yang, S., Yang, H., Jiao, S., and Song, W.L. (2021). Bidirectional planar flexible snake-origami batteries. *Adv. Sci.* **8**, 2101372.
89. Zhang, P., Wang, F., Yu, M., Zhuang, X., and Feng, X. (2018). Two-dimensional materials for miniaturized energy storage devices: from individual devices to smart integrated systems. *Chem. Soc. Rev.* **47**, 7426–7451.
90. Zhai, S., Wang, N., Tan, X., Jiang, K., Quan, Z., Li, Y., and Li, Z. (2021). Interface-Engineered Dendrite-Free Anode and Ultraconductive Cathode for Durable and High-Rate Fiber Zn Dual-Ion Microbattery. *Adv. Funct. Mater.* **31**, 2008894.
91. Wang, K., Zhang, X., Han, J., Zhang, X., Sun, X., Li, C., Liu, W., Li, Q., and Ma, Y. (2018). High-Performance Cable-Type Flexible Rechargeable Zn Battery Based on MnO_2 @CNT Fiber Microelectrode. *ACS Appl. Mater. Interfaces* **10**, 24573–24582.
92. Sun, X., Yang, Y., Liu, Q., Zheng, D., Shao, C., Wang, Y., Lv, J., Yang, T., Lu, Y., Ren, Q., and Chen, N. (2024). Accelerating wound healing with flexible zinc ion microbatteries coupled with endogenous electrical fields. *Nano Energy* **123**, 109425.
93. Chen, J., Zhao, W., Gao, Y., Bu, F., Wu, P., Xiang, Y., Wang, J., Liu, X., and Guan, C. (2025). Single-layer MoS_2 /graphene-based stable on-chip Zn-ion microbattery for monolithically integrated electronics. *Sci. Bull.* **70**, 508–517.
94. Li, M., Guo, Y., Yuan, C., Li, J., Guo, C., Zhang, J., Wang, X., Wang, J., Wang, J., Hu, Z., et al. (2024). 3D printing of customized Li-S microbatteries. *Nano Energy* **130**, 110101.
95. Qu, Z., Ma, J., Huang, Y., Li, T., Tang, H., Wang, X., Liu, S., Zhang, K., Lu, J., Karnaushenko, D.D., et al. (2024). A Photolithographable Electrolyte for Deeply Rechargeable Zn Microbatteries in On-Chip Devices. *Adv. Mater.* **36**, 2310667.
96. Lai, W., Erdonmez, C.K., Marinis, T.F., Bjune, C.K., Dudney, N.J., Xu, F., Wartena, R., and Chiang, Y.M. (2010). Ultrahigh-energy-density microbatteries enabled by new electrode architecture and micropackaging design. *Adv. Mater.* **22**, E139–E144.
97. Zhai, S., Wang, N., Tan, X., Jiang, K., Quan, Z., Li, Y., and Li, Z. (2021). Interface-engineered dendrite-free anode and ultraconductive cathode for durable and high-rate fiber Zn dual-ion microbattery. *Adv. Funct. Mater.* **31**, 2008894.
98. Weng, Q., Wang, S., Liu, L., Lu, X., Zhu, M., Li, Y., Gabler, F., and Schmidt, O.G. (2021). A compact tube-in-tube micro-sized lithium-ion battery as an independent microelectric power supply unit. *Cell Rep. Phys. Sci.* **2**, 100429.
99. Gabler, F., Karnaushenko, D.D., Karnaushenko, D., and Schmidt, O.G. (2019). Magnetic origami creates high performance micro devices. *Nat. Commun.* **10**, 3013.
100. Cao, L., Fang, G., Cao, H., and Duan, X. (2019). Photopatterning and electrochemical energy storage properties of an on-chip organic radical microbattery. *Langmuir* **35**, 16079–16086.
101. Kuang, Y., Chen, C., Kirsch, D., and Hu, L. (2019). Thick electrode batteries: principles, opportunities, and challenges. *Adv. Energy Mater.* **9**, 1901457.
102. Liu, L., Weng, Q., Lu, X., Sun, X., Zhang, L., and Schmidt, O.G. (2017). Advances on micro-sized on-chip lithium-ion batteries. *Small* **13**, 1701847.
103. Yue, X., Johnson, A.C., Kim, S., Kohlmeyer, R.R., Patra, A., Grzyb, J., Padmanabha, A., Wang, M., Jiang, Z., Sun, P., et al. (2021). A Nearly Packaging-Free Design Paradigm for Light, Powerful, and Energy-Dense Primary Microbatteries. *Adv. Mater.* **33**, 2101760.
104. Jin, X., Song, L., Dai, C., Xiao, Y., Han, Y., Li, X., Wang, Y., Zhang, J., Zhao, Y., Zhang, Z., et al. (2022). A Flexible Aqueous Zinc-Iodine Microbattery with Unprecedented Energy Density. *Adv. Mater.* **34**, 2109450.
105. Zhu, Z., Kan, R., Hu, S., He, L., Hong, X., Tang, H., and Luo, W. (2020). Recent Advances in High-Performance Microbatteries: Construction, Application, and Perspective. *Small* **16**, 2003251.
106. Zhu, M., Hu, J., Lu, Q., Dong, H., Karnaushenko, D.D., Becker, C., Karnaushenko, D., Li, Y., Tang, H., Qu, Z., et al. (2021). A patternable and in situ formed polymeric zinc blanket for a reversible zinc anode in a skin-mountable microbattery. *Adv. Mater.* **33**, 2007497.
107. Letiche, M., Eustache, E., Freixas, J., Demortière, A., De Andrade, V., Morgenroth, L., Tilmant, P., Vaurette, F., Troadec, D., and Roussel, P. (2017). Atomic layer deposition of functional layers for on chip 3D Li-ion all solid state microbattery. *Adv. Energy Mater.* **7**, 1601402.
108. Hur, J.I., Smith, L.C., and Dunn, B. (2018). High areal energy density 3D lithium-ion microbatteries. *Joule* **2**, 1187–1201.
109. Chamran, F., Yeh, Y., Min, H.-S., Dunn, B., and Kim, C.-J. (2007). Fabrication of high-aspect-ratio electrode arrays for three-dimensional microbatteries. *J. Microelectromech. Syst.* **16**, 844–852.
110. Li, Y.-Q., Shi, H., Wang, S.-B., Zhou, Y.-T., Wen, Z., Lang, X.-Y., and Jiang, Q. (2019). Dual-phase nanostructuring of layered metal oxides for high-performance aqueous rechargeable potassium ion microbatteries. *Nat. Commun.* **10**, 4292.
111. Liu, H., Zhang, G., Wang, L., Zhang, X., Zhao, Z., Chen, F., Song, L., and Duan, H. (2021). Engineering 3D Architecture Electrodes for High-Rate Aqueous Zn–Mn Microbatteries. *ACS Appl. Energy Mater.* **4**, 10414–10422.
112. Jin, Y., Zou, L., Liu, L., Engelhard, M.H., Patel, R.L., Nie, Z., Han, K.S., Shao, Y., Wang, C., Zhu, J., et al. (2019). Joint Charge Storage for High-Rate Aqueous Zinc-Manganese Dioxide Batteries. *Adv. Mater.* **31**, e1900567.
113. Qu, Z., Zhu, M., Yin, Y., Huang, Y., Tang, H., Ge, J., Li, Y., Karnaushenko, D.D., Karnaushenko, D., and Schmidt, O.G. (2022). A Sub-Square-Millimeter Microbattery with Milliampere-Hour-Level Footprint Capacity. *Adv. Energy Mater.* **12**, 2200714.
114. Zhao, B., Ma, W., Li, B., Hu, X., Lu, S., Liu, X., Jiang, Y., and Zhang, J. (2022). A fast and low-cost interface modification method to achieve high-performance garnet-based solid-state lithium metal batteries. *Nano Energy* **91**, 106643.
115. Pan, J., Zhao, P., Wang, N., Huang, F., and Dou, S. (2022). Research progress in stable interfacial constructions between composite polymer electrolytes and electrodes. *Energy Environ. Sci.* **15**, 2753–2775.
116. Jiang, C., Fang, Y., Lang, J., and Tang, Y. (2017). Integrated Configuration Design for Ultrafast Rechargeable Dual-Ion Battery. *Adv. Energy Mater.* **7**, 1700913.
117. Zhao, W., Wang, H., Li, S., Tang, X., Zhang, M., Mustehsin, A., Bai, M., Liu, S., Sun, C., and Ma, Y. (2021). A flexible, integrated film battery configuration for ultrafast sodium ion storage. *J. Mater. Chem. A* **9**, 1252–1259.
118. Wang, A., Zhou, W., Chen, M., Huang, A., Tian, Q., Xu, X., and Chen, J. (2021). Integrated design of aqueous zinc-ion batteries based on dendrite-free zinc microspheres/carbon nanotubes/nanocellulose composite film anode. *J. Colloid Interface Sci.* **594**, 389–397.
119. Zhang, Y., Wang, Q., Bi, S., Yao, M., Wan, F., and Niu, Z. (2019). Flexible all-in-one zinc-ion batteries. *Nanoscale* **11**, 17630–17636.
120. Lobe, S., Bauer, A., Uhlenbruck, S., and Fattakhova-Rohlfing, D. (2021). Physical Vapor Deposition in Solid-State Battery Development: From Materials to Devices. *Adv. Sci.* **8**, 2002044.
121. Wang, Y., Jiang, H., Zheng, R., Pan, J., Niu, J., Zou, X., and Jia, C. (2020). A flexible, electrochromic, rechargeable Zn-ion battery based on actiniae-like self-doped polyaniline cathode. *J. Mater. Chem. A* **8**, 12799–12809.
122. Yao, M., Yuan, Z., Li, S., He, T., Wang, R., Yuan, M., and Niu, Z. (2021). Scalable Assembly of Flexible Ultrathin All-in-One Zinc-Ion Batteries with Highly Stretchable, Editable, and Customizable Functions. *Adv. Mater.* **33**, 2008140.

123. Chen, P., Zhou, W., Xiao, Z., Li, S., Wang, Z., Wang, Y., and Xie, S. (2020). An integrated configuration with robust interfacial contact for durable and flexible zinc ion batteries. *Nano Energy* 74, 104905.
124. Ekstedt, S., Wysocki, M., and Asp, L.E. (2010). Structural batteries made from fibre reinforced composites. *Plast. Rubber Compos.* 39, 148–150.
125. Katsuyama, Y., Hui, J., Thiel, M., Haba, N., Yang, Z., and Kaner, R.B. (2024). 3D-Printed Carbon Scaffold for Structural Lithium Metal Batteries. *Small Methods* 8, 2400831.
126. Han, Q., Zhang, W., Han, Z., Wang, F., Geng, D., Li, X., Li, Y., and Zhang, X. (2019). Preparation of PAN-based carbon fiber@MnO₂ composite as an anode material for structural lithium-ion batteries. *J. Mater. Sci.* 54, 11972–11982.
127. Chen, J., Zhou, Y., Islam, M.S., Cheng, X., Brown, S.A., Han, Z., Rider, A. N., and Wang, C.H. (2021). Carbon fiber reinforced Zn–MnO₂ structural composite batteries. *Compos. Sci. Technol.* 209, 108787.
128. Kjell, M.H., Jacques, E., Zenkert, D., Behm, M., and Lindbergh, G. (2011). PAN-Based Carbon Fiber Negative Electrodes for Structural Lithium-Ion Batteries. *J. Electrochem. Soc.* 158, A1455–A1460.
129. Pattarakunnan, K., Galos, J., Tian, W., Best, A.S., Kyratzis, I.L., and Mouritz, A.P. (2024). Permeability and barrier layer properties of a woven carbon fibre polymer composite as battery packaging. *Compos. Part C: Open Access* 14, 100490.
130. Lim, G.J.H., Koh, J.J., Chan, K.K., Verma, V., Chua, R., Koh, X.Q., Kidkhunthod, P., Sutrisnoh, N.A.A., and Srinivasan, M. (2024). Amorphous Cellulose Electrolyte for Long Life and Mechanically Robust Aqueous Structural Batteries. *Adv. Funct. Mater.* 34, 202313531.
131. Han, J., Lee, M.J., Lee, K., Lee, Y.J., Kwon, S.H., Min, J.H., Lee, E., Lee, W., Lee, S.W., and Kim, B.J. (2023). Role of Bicontinuous Structure in Elastomeric Electrolytes for High-Energy Solid-State Lithium-Metal Batteries. *Adv. Mater.* 35, 2205194.
132. Deshpande, S., Vidyaprakash, V., Oka, S., Dasari, S.S., Liu, K.W., Wang, C., Lutkenhaus, J.L., and Green, M.J. (2024). Low-Temperature Structural Battery Electrolytes Produced by Polymerization-Induced Phase Separation. *ACS Appl. Polym. Mater.* 6, 6323–6333.
133. Liu, Z., Hu, Q., Guo, S., Yu, L., and Hu, X. (2021). Thermoregulating separators based on phase-change materials for safe lithium-ion batteries. *Adv. Mater.* 33, 2008088.
134. Silva, G., Assis Dutra, T., Nunes-Pereira, J., and Silva, A.P. (2024). Coupled and decoupled structural batteries: A comparative analysis. *J. Power Sources* 604, 234392.
135. Ma, X., Bao, Y., Li, N., Lu, B., Song, Y., Zhang, J., and Fang, D. (2024). Customizable 3D-printed decoupled structural lithium-ion batteries with stable cyclability and mechanical robustness. *Compos. Sci. Technol.* 256, 110783.
136. Nagasubramanian, G., Jungst, R.G., and Doughty, D.H. (1999). Impedance, power, energy, and pulse performance characteristics of small commercial Li-ion cells. *J. Power Sources* 83, 193–203.
137. Zhou, R. (2023). Analysis of Tesla Model's Cylindrical Cells and Electric Motor. *Highlights Sci. Eng. Technol.* 43, 490–498.
138. Yang, X.-G., Liu, T., and Wang, C.-Y. (2021). Thermally modulated lithium iron phosphate batteries for mass-market electric vehicles. *Nat. Energy* 6, 176–185.
139. Fichtner, M. (2022). Recent research and progress in batteries for electric vehicles. *Batter. Supercaps* 5, e202100224.
140. Yang, Y., Dorn-Gomba, L., Rodriguez, R., Mak, C., and Emadi, A. (2020). Automotive power module packaging: Current status and future trends. *IEEE Access* 8, 160126–160144.
141. Chen, A., Guo, X., Yang, S., Liang, G., Li, Q., Chen, Z., Huang, Z., Yang, Q., Han, C., and Zhi, C. (2021). Human joint-inspired structural design for a bendable/foldable/stretchable/twistable battery: achieving multiple deformabilities. *Energy Environ. Sci.* 14, 3599–3608.
142. Kwon, Y.H., Woo, S.W., Jung, H.R., Yu, H.K., Kim, K., Oh, B.H., Ahn, S., Lee, S.Y., Song, S.W., Cho, J., et al. (2012). Cable-type flexible lithium ion battery based on hollow multi-helix electrodes. *Adv. Mater.* 24, 5192–5197.
143. Chen, T., Qiu, L., Cai, Z., Gong, F., Yang, Z., Wang, Z., and Peng, H. (2012). Intertwined aligned carbon nanotube fiber based dye-sensitized solar cells. *Nano Lett.* 12, 2568–2572.
144. Mo, F., Liang, G., Huang, Z., Li, H., Wang, D., and Zhi, C. (2020). An overview of fiber-shaped batteries with a focus on multifunctionality, scalability, and technical difficulties. *Adv. Mater.* 32, 1902151.
145. Zhang, K., Shi, X., Jiang, H., Zeng, K., Zhou, Z., Zhai, P., Zhang, L., and Peng, H. (2024). Design and fabrication of wearable electronic textiles using twisted fiber-based threads. *Nat. Protoc.* 19, 1557–1589. <https://doi.org/10.1038/s41596-024-00956-6>.
146. Liu, G., Zhang, X.-y., Lu, B., Song, Y., Qiao, Y., Guo, X., Ao, S., Zhang, J., Fang, D., and Bao, Y. (2022). Crocodile skin inspired rigid-supple integrated flexible lithium ion batteries with high energy density and bidirectional deformability. *Energy Storage Mater.* 47, 149–157.
147. Aubin, C.A., Choudhury, S., Jerch, R., Archer, L.A., Pikul, J.H., and Shepherd, R.F. (2019). Electrolytic vascular systems for energy-dense robots. *Nature* 571, 51–57.
148. Li, T., Li, G., Liang, Y., Cheng, T., Dai, J., Yang, X., Liu, B., Zeng, Z., Huang, Z., Luo, Y., et al. (2017). Fast-moving soft electronic fish. *Sci. Adv.* 3, e1602045.
149. Tang, K., and Ren, Y. (2022). Practical Application of Room Temperature Na-Ion Batteries. *Sodium-Ion Batteries. Materials, Characterization, and Technology* 2, 531–549.
150. Zhang, Y., Ma, J., Singh, A.K., Cao, L., Seo, J., Rahn, C.D., Bakis, C.E., and Hickner, M.A. (2017). Multifunctional structural lithium-ion battery for electric vehicles. *J. Intell. Mater. Syst. Struct.* 28, 1603–1613.
151. Thomas, J.P., Keenon, M.T., DuPasquier, A., Qidwai, M.A., and Matic, P. (2003). Multifunctional structure-battery materials for enhanced performance in small unmanned air vehicles. In *Proceedings of IMECE2003: International Mechanical Engineering Congress and Exposition (ASME)*, pp. 289–292.
152. Kühnelt, H., Beutl, A., Mastropiero, F., Laurin, F., Willrodt, S., Bismarck, A., Guida, M., and Romano, F. (2021). Structural Batteries for Aeronautic Applications—State of the Art, Research Gaps and Technology Development Needs. *Aerospace* 9, 7.
153. Karadotcheva, E., Nguyen, S.N., Greenhalgh, E.S., Shaffer, M.S.P., Kucernak, A.R.J., and Linde, P. (2021). Structural power performance targets for future electric aircraft. *Energies* 14, 6006.
154. Nguyen, S.N., Millereux, A., Pouyat, A., Greenhalgh, E.S., Shaffer, M.S.P., Kucernak, A.R.J., and Linde, P. (2021). Conceptual multifunctional design, feasibility and requirements for structural power in aircraft cabins. *J. Aircr.* 58, 677–687.
155. Rothgang, S., Baumhöfer, T., van Hoek, H., Lange, T., De Doncker, R.W., and Sauer, D.U. (2015). Modular battery design for reliable, flexible and multi-technology energy storage systems. *Appl. Energy* 137, 931–937.
156. Cicconi, P., Kumar, P., and Varshney, P. (2020). A support approach for the modular design of Li-ion batteries: A test case with PCM. *J. Energy Storage* 31, 101684.
157. Chen, H., Fan, J., Zhang, M., Feng, X., Zhong, X., He, J., and Ai, S. (2023). Mechanism of inhomogeneous deformation and equal-stiffness design of large-format prismatic lithium-ion batteries. *Appl. Energy* 332, 120494.
158. Talele, V., Patil, M.S., Panchal, S., Fraser, R., Fowler, M., and Gunti, S.R. (2023). Novel metallic separator coupled composite phase change material passive thermal design for large format prismatic battery pack. *J. Energy Storage* 58, 106336.
159. Wu, B., Li, Z., and Zhang, J. (2014). Thermal design for the pouch-type large-format lithium-ion batteries: I. Thermo-electrical modeling and origins of temperature non-uniformity. *J. Electrochem. Soc.* 162, A181–A191.
160. Jin, C., Sun, Y., Yao, J., Feng, X., Lai, X., Shen, K., Wang, H., Rui, X., Xu, C., Zheng, Y., et al. (2022). No thermal runaway propagation optimization

- design of battery arrangement for cell-to-chassis technology. *ETransportation* **14**, 100199.
161. Lin, J., Zhang, X., Fan, E., Chen, R., Wu, F., and Li, L. (2023). Carbon neutrality strategies for sustainable batteries: from structure, recycling, and properties to applications. *Energy Environ. Sci.* **16**, 745–791.
162. Yubo Lian, B.Y., Cui, Y., Tian, H., Yan, J., and Chen, C. (2023). Research on Integrated Design of Battery Pack and Car Body Based on Torsional Stiffness. *Automot. Eng.* **45**, 647–653.
163. Goldberg, B., Zufferey, R., Doshi, N., Helbling, E.F., Whittredge, G., Kovac, M., and Wood, R.J. (2018). Power and control autonomy for high-speed locomotion with an insect-scale legged robot. *IEEE Robot. Autom. Lett.* **3**, 987–993.
164. Xia, Q., Zan, F., Zhang, Q., Liu, W., Li, Q., He, Y., Hua, J., Liu, J., Xu, J., Wang, J., et al. (2023). All-solid-state thin film lithium/lithium-ion micro-batteries for powering the Internet of things. *Adv. Mater.* **35**, 2200538.
165. Chen, S., Wu, G., Jiang, H., Wang, J., Chen, T., Han, C., Wang, W., Yang, R., Zhao, J., Tang, Z., et al. (2025). External Li supply reshapes Li deficiency and lifetime limit of batteries. *Nature* **638**, 676–683.
166. Bradford, G., Lopez, J., Ruza, J., Stolberg, M.A., Osterude, R., Johnson, J.A., Gomez-Bombarelli, R., and Shao-Horn, Y. (2023). Chemistry-informed machine learning for polymer electrolyte discovery. *ACS Cent. Sci.* **9**, 206–216.
167. Zhang, Z., Tian, Y., Liu, Z., Wang, W., Ni, Y., Zhang, D., and Tang, H. (2024). Molecular dynamics of monocrystal copper nano-scratched with tungsten tip. *Mater. Today Commun.* **40**, 109714.
168. Li, A., Yuen, A.C.Y., Wang, W., Chen, T.B.Y., Lai, C.S., Yang, W., Wu, W., Chan, Q.N., Kook, S., and Yeoh, G.H. (2022). Integration of computational fluid dynamics and artificial neural network for optimization design of battery thermal management system. *Batteries* **8**, 69.
169. Wang, J., Feng, X., Yu, Y., Huang, H., Zheng, M., Xu, Y., Wu, J., Yang, Y., and Lu, J. (2024). Rapid temperature-responsive thermal regulator for safety management of battery modules. *Nat. Energy* **9**, 939–946.
170. Ma, S., Xue, P., Valenzuela, C., Liu, Y., Chen, Y., Feng, Y., Bi, R., Yang, X., Yang, Y., Sun, C., et al. (2025). 4D-Printed Adaptive and Programmable Shape-Morphing Batteries. *Adv. Mater.* **37**, 2505018.
171. Jin, T., Ma, Y., Xiong, Z., Fan, X., Luo, Y., Hui, Z., Chen, X., and Yang, Y. (2021). Bioinspired, Tree-Root-Like Interfacial Designs for Structural Batteries with Enhanced Mechanical Properties. *Adv. Energy Mater.* **11**, 2100997.